

The Sanitation System Management Standard: Part 1 — Asset Management

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Foreword

This document was developed by the mWater Foundation. mWater Foundation is an international non-governmental organization based in the United States that is dedicated to developing data-driven approaches for improving access to safe water and sanitation globally. The document draws on more than 14 years of experience by mWater in working with many government and civil society partners to develop digital tools and standardized approaches for managing data about water and sanitation systems in low-resource areas of the world. This Sanitation Asset Standard builds on the success of the Water Asset Standard, also developed by mWater in 2022.

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Introduction

This document was developed in response to demand from sanitation practitioners and development partners to specify a data structure and useful set of definitions to facilitate the capture, use, and sharing of data about sanitation infrastructure.

Sanitation infrastructure includes the physical assets involved in the containment, collection, conveyance, treatment, and safe disposal or reuse of wastewater and faecal sludge. Physical assets of a sanitation system include hydraulic structures and components, electromechanical equipment, and civil engineering works, and span both sewered (network-based) and non-sewered (on-site) sanitation.

Classification of assets

An asset management system tracks the key data about each asset that an organization considers valuable or important enough to actively manage. An organization must balance the need for data about assets with the effort involved in collecting data and keeping it up to date. There is no single right answer to the question of which assets need to be managed or tracked,

since this depends on the needs and capabilities of the management organization. For this reason, this document does not mandate which asset types or attributes need to be used in an asset management system. To facilitate the development of an asset management system, this standard provides definitions for the most common types of assets found in sanitation systems.

This standard classifies assets using a two-tiered structure that consists of **types** of assets that are organized into **classes**. The asset types are unique and sufficient to describe an asset. Classes are not required to define a type, they are simply useful groupings that reflect common industry practices. The standard currently recognizes 27 types of assets and 4 classes, as shown in Figure 1.

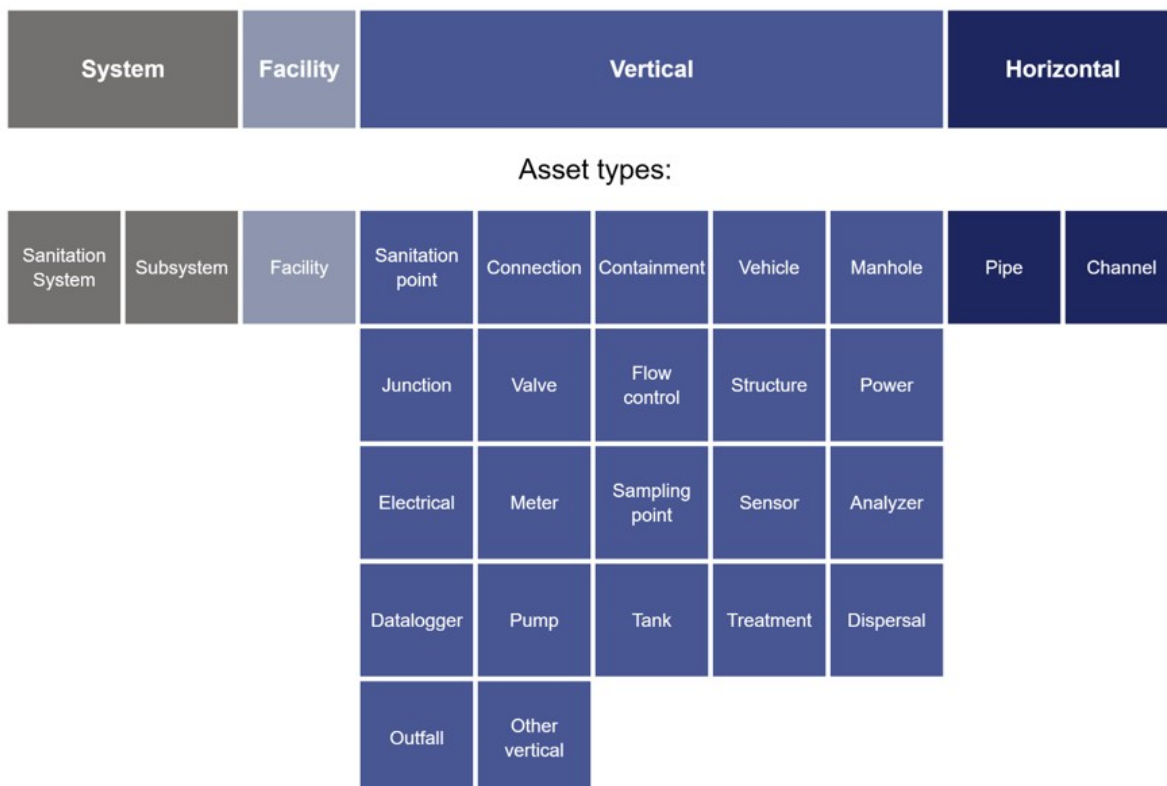


Figure 1: Asset classes and types

Attributes

Attributes are the individual pieces of data about a particular asset that describe important characteristics, such as the equipment manufacturer and model, condition and current operational status, and other technical details that relate to certain types of assets.

Any decision to include an attribute in a standard involves trade-offs between the need for comprehensive data and the practical challenges involved in collecting and updating the data. The following criteria were considered when evaluating potential attributes for this standard:

- **Is the attribute asset data?** Asset data pertains to the infrastructure and equipment used to provide services. Data on the quality of services provided, operations and maintenance, effluent quality, and customer service are all important for effective asset management, but these data are usually kept in other data systems.

- **Is the attribute in widespread use?** More consideration was given to defining attributes that are commonly used by sanitation service providers, engineers, and technicians. This helps to determine the relative importance of collecting the data.
- **Is the attribute used to make decisions?** The focus of this standard is on data use by the service provider, as well as longer term data use for planning and financial modeling. Attributes that are needed to make specific decisions were emphasized over those that were more descriptive in nature.

The standard recognizes that there are some **general attributes** that apply to most or all assets and other **type-specific attributes** that only are relevant for certain kinds of assets. For example, most (but not all) types of equipment have a manufacturer, model, and serial number, but only containment structures have a storage volume.

Distinguishing between general and type-specific attributes is helpful but not always sufficient to determine which kind of attribute applies to a certain asset, so the standard provides two additional concepts to help manage attributes:

- **Groups** are headings that organize data and keep similar attributes together. Every attribute in this standard has a group, even if there is only one attribute in the group. This is intended to provide consistency for programmers to design user interfaces that work the same way for any asset.
- **Conditions** are rules, based on the values of other attributes or the type of asset, that determine whether or not an attribute is applicable to a particular asset. For example, if the shape of a tank is cylindrical, then the diameter and height need to be recorded whereas a rectangular tank has a length and width but no diameter. Conditions may be applied to individual attributes or to groups of attributes.

The data model for attributes described in this section is summarized in Figure 2.



Figure 2: Illustration of general and type-specific attributes and groups of attributes

Organization of assets

Asset management systems typically use hierarchical classification schemes and parent-child relationships to help keep assets organized and help field staff to quickly identify or locate an asset. This standard accommodates up to 7 optional levels of hierarchical classification using attributes whose values are determined and enforced by the organization implementing the standard. In addition, any asset can be assigned a parent asset, creating a parent-child relationship between the two assets.

Standardized names and definitions

All types and attributes in this standard have unique names to improve clarity and interoperability in asset management systems. In addition, attributes have a unique numerical ID code that specifies the class (1-digit), type (2-digits), and attribute (2-digits). In cases where attributes have a fixed set of possible values, the choices are also given names and are further defined by descriptions. The descriptions and definitions in this document are based on industry standard terms and cross-referenced against relevant standards and recognized sanitation-sector references. Implementers are free to substitute their own terms or translations for the types, attributes, and choices specified in this standard but they should ensure that they are traceable back to the standard term for the purposes of data exchange and publication.

Using the standard

Organizations are encouraged to use this standard to design and develop infrastructure asset management systems that meet their unique needs, capabilities, and values. The standard is simple enough that it could be implemented using a spreadsheet or ledger, but organizations are encouraged to incorporate the standard into geographical information systems (GIS), online database systems, and mobile apps.

1 Scope

This document provides guidance for managing and sharing data about a sanitation system.

This standard is applicable to any engineered or constructed works designed to contain, collect, convey, treat, and safely dispose of or reuse wastewater and faecal sludge, spanning both sewerage (network-based) and non-sewerage (on-site) sanitation. Use of this standard does not require or depend on the use of a particular software application or database technology.

The scope of this document is limited to providing a standard description of the structure and types of data required to define the configuration, condition, and performance of the physical assets of a sanitation system. Readers are encouraged to consult other standards and resources before implementing an infrastructure asset management process, including ISO standards 55000, 55001, and 55002, related to the asset management process, and ISO 24516, which provides specific guidance for the management of assets of water supply and wastewater systems.

2 Normative references

There are no normative references in this document.

A normative reference is an outside standard that is required in order to use or interpret the current document. In the current version of this standard, normative references are not necessary.

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

3.0.1 asset ID

a code used to identify a particular asset that is unique within the asset register (see Section 5.2 for requirements regarding unique identification of assets)

3.0.2 asset management system

the policies and procedures that an organization uses to manage assets

3.0.3 asset portfolio

assets that are within the scope of the asset management system

3.0.4 asset register

a complete listing of the asset information of an organization

3.0.5 attribute ID

a unique 5-digit code used in this standard to refer to a specific attribute

3.0.6 conduit

any artificial or natural duct, either open or closed, for conveying fluids

3.0.7 faecal sludge

the accumulated excreta and wastewater stored in on-site containment technologies that are removed by emptying rather than carried away by a sewer network

3.0.8 sanitation system

the combined infrastructure and services that contain, collect, convey, treat, and safely dispose of or reuse wastewater and faecal sludge

3.0.9 septage

the mixture of settled solids and liquid removed from a septic tank or similar on-site containment technology when it is emptied (desludged)

3.0.10 unit of measurement

a definite magnitude of a physical quantity, defined by convention or law, that is used as a standard for measurement of the same kind of quantity

3.0.11 wastewater

water whose quality has been degraded by domestic, commercial, industrial, or institutional use, together with the excreta and other substances it carries

4 Data types and units of measurement

Data types are used by computer programmers to help define the possible values a particular piece of data is allowed to have and the types of operations or transformations that can be performed on the data. This standard includes the data type for each attribute in Section 4.1 to help with the design of databases for asset data and to facilitate the transfer of asset data from one system to another. In cases where an attribute involves a physical measurement of some well-defined quantity, the standard also defines the units of measurement in Section 4.2, as well as the conversion factors between different units.

4.1 Data types

The following table describes the minimum data types needed to use this standard. In order to promote understanding among sanitation professionals who might not be familiar with computer engineering terms, this section uses non-technical language to specify data types.

Table 1. Data types

Data type	Description
Text	Any combination of letters, numbers, punctuation, and spaces (also known as Character in some programming languages)
Choice	Text field that has a fixed set of possible values (also known as an Enumeration or Factor)
Number	Any integer or decimal number, consisting of a whole number part and an optional decimal part, with either a fixed or variable number of digits after the decimal point
Date	Calendar date according to the Gregorian calendar that includes the 4-digit year, 2-digit month, and 2-digit day in that order, and optionally separated by a delimiting character
Geometry	Location of a point, line, or polygon on the surface of the earth, according to the WGS84 geographic coordinate system (any common file format for geographical data may be used, including GeoJSON or ESRI Shape File)
Checkbox	Binary outcome that can only have values of TRUE or FALSE; checkbox questions default to FALSE unless the user changes the value to TRUE (also known as Boolean or Logical)
Image	Image file in any common format, such as JPG or PNG
Unit	Physical quantity that has both a magnitude (a number) and a unit of measurement

4.2 Units of measurement

Attributes that have the data type of Unit (see Section 4.1) include the name of the physical quantity that is being measured. The name "Unit" is followed by a colon (":") and then the name of the physical quantity. For example, the physical quantity of Length would appear in the Data type column of the attribute tables as: "Unit: Length".

The units of measurement required to use this standard and their conversion factors are defined in Table 2. Each physical quantity has a base unit, which appears first in the table, and is defined by the International System of Units (SI). Conversion factors included in this section for SI units are derived from the SI Brochure (BIPM 2019).

The conversion factors are expressed with reference to the base unit, such that within each physical quantity the conversion factor for the base unit is always equal to 1. This means that a conversion factor can be treated as the ratio of the current unit to the base unit, or the amount of the base unit that would be equal to one current unit. In the case of temperature, the conversion factor approach does not apply; instead, follow the conversion formulas provided in Table 2.

Table 2. Units of measurement

Physical quantity / Symbol	Name	Conversion factor	System
Length			
m	meters	1	SI
cm	centimeters	0.01	SI
mm	millimeters	0.001	SI
µm	micrometers	1×10^{-6}	SI
nm	nanometers	1×10^{-9}	SI
km	kilometers	1000	SI
in	inches	0.0254	US
ft	feet	0.3048	US

Physical quantity / Symbol	Name	Conversion factor	System
yd	yards	0.9144	US
mi	miles	1609.344	US
Pressure			
Pa	pascals	1	SI
kPa	kilopascals	1000	SI
psi	pounds per square inch	6894.757	US
ft H ₂ O	feet of water (39.2 °F)	2988.98	US
in H ₂ O	inches of water (39.2 °F)	249.082	US
m H ₂ O	meters of water (4 °C)	9806.65	SI
cm H ₂ O	centimeters of water (4 °C)	98.0665	SI
mm Hg	millimeters of mercury (32 °F)	133.3224	SI
in Hg	inches of mercury (32 °F)	3386.38	US
atm	atmospheres	101325	SI
bar	bar	100000	SI
Flow rate			
m ³ /s	cubic meters per second	1	SI
m ³ /d	cubic meters per day	8.64 X 10 ⁴	SI
m ³ /hr	cubic meters per hour	3600	SI
L/s	liters per second	0.001	SI
L/min	liters per minute	60	SI
L/d	liters per day	86.4	SI
ML/d	million liters per day	8.64 X 10 ⁷	SI
gal/min (gpm)	gallons per minute	6.30902 X 10 ⁵	US
gal/d	gallons per day	4.38126 X 10 ⁸	US
Mgal/d (mgd)	million gallons per day	0.0438126	US
Mass flow			
kg/d	kilograms per day	1	SI
lb/d	pounds per day	0.453592	US
Area			
m ²	square meters	1	SI
ha	hectares	10000	SI
ft ²	square feet	0.09290304	US
acre	acres	4046.8564	US
Voltage			
V	volts	1	SI
kV	kilovolts	1000	SI
MV	megavolts	1 X 10 ⁶	SI

Physical quantity / Symbol	Name	Conversion factor	System
Power			
W	watts	1	SI
kW	kilowatts	1000	SI
MW	megawatts	1×10^6	SI
hp	horsepower	746	US
kVA	kilovolt-amps	1000	SI
Current			
A	amperes	1	SI
Volume			
m ³	cubic meters	1	SI
L	liters	0.001	SI
ft ³	cubic feet	0.02831685	US
gal	gallon	0.003785412	US
Mm ³	million cubic meters	1×10^6	SI
Mgal	million gallons	3785	US
acre-ft	acre-feet	1223.489	US
Time			
d	days	1	SI
mo	months	30.436875	SI
yr	years	365.2425	SI
Temperature			
°C	degrees celsius	$t_F = t_C \times 1.8 + 32$	SI
°F	degrees fahrenheit	$t_C = (t_F - 32) / 1.8$	US

* Temperature is a special case. Use the formula provided, where t_F = temperature in degrees Fahrenheit and t_C = temperature in degrees Celsius.

5 Assets

5.1 Definition of an asset

An asset is an item, thing, or entity that has potential or actual value to an organization (ISO 55000). Sanitation systems generally consist of physical assets, which are engineered, manufactured, or constructed, spanning both sewerage (network-based) and non-sewerage (on-site) sanitation. For the purposes of database design, an asset consists of one single row in the table of assets.

5.2 Unique identification of assets

All assets shall be assigned an identification code, referred to as the Asset ID (section 6.4), that is unique with respect to all other assets included in the asset register. Identification codes may

be text strings, numerical codes, or mixtures of text and numbers. In order to facilitate the tracking of maintenance interventions and failures over time, the asset ID shall be immutable, meaning it shall never be changed, even when an asset is moved or updated, and it shall never be reused when an asset is replaced or removed from service.

Organizational data managers and software providers should consider developing procedures or algorithms to check for and resolve duplicate asset IDs, especially when assets are created offline and later uploaded to a central database.

An organization may choose to use a hierarchical coding system, in which parts of the asset ID refer to types, facilities, or other categories that assist in identifying the location or function of the asset. Organizations setting up hierarchical coding systems should take into consideration the following:

- Assets are often moved from one location to another;
- Names of administrative regions and facilities change from time to time; and
- The asset owner or managing organization might change for a facility.

The asset coding system and staff procedures shall be designed to maintain a unique and immutable Asset ID over the entire service life of the asset.

5.3 Parent and child relationships

Any asset shall have an optional parent asset, which is defined by the value of the Parent ID attribute of the child asset. A child asset shall have only one parent asset, but parent assets may have an unlimited number of child assets.

5.4 Asset classes

Asset classes are useful groupings of asset types that fulfill similar functions and tend to have a common asset management strategy (AWWA 2018). In this document, asset classes are not necessary to include when describing an asset because the asset types (next section) are unique and sufficient to define the required attributes. However, the asset class can be useful for organizing lists of asset types during data collection, searching, and analysis.

The asset classes provided in the following table shall be accommodated in asset management systems.

Table 3. Asset classes

Asset class	Description	Geometry
System	Collection of assets that are managed in a similar manner to provide a service.	Point or polygon
Facility	Physical location where specific functions are performed.	Point or polygon
Vertical	Asset within a building or facility, or free-standing, often composed of multiple components; also known as an above ground asset (AWWA 2018).	Point
Horizontal	Asset which may be configured or networked for the purpose of moving materials or services from one place to another, such as pipes or conduits (AWWA 2018); also known as below ground assets, but may exist above or below ground.	Line

5.5 Asset types

An asset type is a grouping of assets with common characteristics that distinguish them as a group (ISO 55000). In this document, asset types are used to determine which type-specific attributes can be added to that particular asset. The asset types described in the following tables shall be made available to include in the asset management system.

Table 4. Asset types

Asset class	#	Asset type	Description
System	101	Sanitation system	Collection of assets that work together as a system to contain, collect, transport, treat, and safely dispose of or reuse wastewater and faecal sludge
	102	Subsystem	Functional or administrative subdivision of a larger sanitation system
Facility	201	Facility	Physical location where specific functions are performed in the collection, transport, treatment, disposal, or reuse of wastewater and faecal sludge; optional, used to organize or locate the assets contained in large and complex systems.
Vertical	315	Sanitation point	Location or facility where users access sanitation services, including toilets, latrines, urinals, and communal or public sanitation blocks
	322	Connection	Property or customer location that receives sanitation service
	316	Containment	Structure that holds excreta and wastewater at or near where they are generated; some types also provide partial treatment
	317	Vehicle	Mobile asset used to empty, transport, or transfer faecal sludge, septage, or wastewater
	318	Manhole	Structure that provides access to a sewer for inspection, cleaning, and maintenance; includes inspection chambers and cleanouts
	310	Junction	Section of a piped network that joins two or more pipes or conduits
	308	Valve	Mechanical device designed to close off or regulate the flow of wastewater, sludge, air, or other fluids in a pipe or closed conduit
	321	Flow control	Structure or device that diverts, splits, restricts, releases, or measures flow in an open channel or basin
	314	Structure	Anything that is constructed or built with a fixed location on the ground
	304	Power	Device that supplies electrical power for use by electrical equipment, such as a generator, solar power system, wind turbine, or connection to a power grid
	307	Electrical	Electrical or electronic device
	306	Meter	Device for measuring the flow rate of wastewater, effluent, or sludge through a pipe or open channel
	311	Sampling point	Location or access point that allows a sample of influent, effluent, sludge, or receiving water to be collected for testing
	312	Sensor	Device that measures a physical condition or parameter of interest
	313	Analyzer	Device that conducts periodic or continuous measurements of a wastewater, effluent, or process quality parameter

Asset class	#	Asset type	Description
	323	Datalogger	Device that automatically records measurements over time from connected sensors, meters, or analyzers, and stores the data locally or transmits it to a remote system
	302	Pump	Mechanical or electromechanical device that increases the hydraulic head (pressure) of a fluid flowing through it
	303	Tank	Structure or container used to hold wastewater, sludge, chemicals, or other fluids within a conveyance network or treatment facility; on-site vessels such as septic tanks are classified under Containment
	305	Treatment	Equipment or process unit that improves the physical, chemical, or biological quality of wastewater or faecal sludge, including separation, stabilization, and dewatering of solids
	319	Dispersal	Structure that discharges treated or partially treated effluent to the soil for infiltration and/or evapotranspiration
	320	Outfall	Point or structure where effluent, overflow, or emergency discharge leaves the system and enters a receiving water body or the environment
	399	Other vertical	Any other type of vertical asset not defined elsewhere in this standard
Horizontal	401	Pipe	Conduit that carries wastewater, sludge, or other fluids from one location to another
	402	Channel	Artificial open or covered channel constructed to convey wastewater, effluent, or combined flows from one location to another. Includes natural watercourses that have been substantially modified (lined, regraded, or walled) and are managed as part of the system

6 Attributes

6.1 Definition of an attribute

An attribute is a piece of data that describes a quality or characteristic of an individual asset.

6.2 Characteristics of attributes

Attributes and their usage shall conform to the following characteristics:

1. **Attributes whose values are unknown or not applicable shall be left blank.** The standard does not distinguish between a missing value and an unknown value, but implementers may choose to track additional metadata about attributes that are not answered, such as when an inspector was unable to determine the value.
2. **Attributes with conditions shall only be assigned a value when the condition is satisfied.** Some attributes are only applicable when another attribute has a certain value or when the asset has a specific class or type. These conditions, which may apply to individual attributes or groups of attributes, are provided in the attribute tables.

6.3 How to read and interpret attribute tables

All attributes defined in this standard are organized into tables that follow a consistent format with the following headings:

- **Code:** The 5-digit unique code assigned to each attribute.
- **Group, Attribute, or Choice Name:** The unique name used in the standard to refer to the attribute, group, or choice option. Note that the text formatting defines which of these three kinds of name is being defined:
 - **Group** names are in *bold font* and do not have an associated attribute ID.
 - Attribute names are in normal font and always have an attribute ID in the row where they appear.
 - Choice names are the names of the answer choices available for the attribute immediately above them and are displayed indented and in *italic font*.
- **Description:** A brief phrase or sentence to further define the meaning or use of the attribute or answer choice.
- **Data type:** The data type and unit of measurement defined (see Section 4) to be used with this attribute. In cases where more than one data type is permitted, types are separated by commas (,).
- **Conditions:** The condition that must all be met in order for the group or attribute to be applicable. Conditions are formatted with the name of the attribute, type, or class that the condition depends on, followed by a colon (:) and the values or answer choices that satisfy the condition. In cases where multiple values could satisfy the condition, each value is separated by a comma (,). Note that when a condition appears on the same row as a Group name, it applies to all attributes in the group (meaning all attributes that appear below the group name).

6.4 General attributes

The following attributes shall be available for all assets.

Table 5. General attributes for all assets

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Basic Info			
00001	Type	Type of asset	Choice	
00002	Name	Name used by technical staff to refer to the asset; might be left blank for many lower level assets	Text	
00003	Unique ID	Unique number automatically assigned when creating an asset	Text	
00004	Description	Additional information to help staff locate or identify an asset	Text	
00005	Photo	Photos of the asset	Image	
00006	Parent asset	Asset ID of the asset that is a parent to this asset	Asset ID	

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Location			
00011	Location	Geographic location of the asset on the surface of the Earth; may be a point, line, or polygon	Geometry	
00012	Location precision	Estimated uncertainty in the geographic coordinates, expressed as a distance	Unit: Length	
00013	Survey elevation	Height above a reference surface, as measured by surveying or high-precision GNSS equipment; used for hydraulic modeling and engineering design. This value is distinct from the altitude captured automatically with GPS coordinates. For chambered assets such as manholes, the surveyed point is the center of the cover	Unit: Length	
00024	Survey elevation datum	Vertical reference system used for Survey elevation and Invert level values	Choice	
	<i>Ellipsoid (WGS84)</i>	Height above the WGS84 reference ellipsoid, as reported by GNSS receivers		
	<i>Mean sea level</i>	Height above local mean sea level or a national vertical datum		
	<i>Other</i>	Any other vertical reference system; describe in Additional technical details		
00014	Elevation precision	Estimated uncertainty in the measurement of the elevation	Unit: Length	
00015	Administrative region	Name that describes the administrative geography of the location; multiple levels may be separated by commas starting with the most general area, for example: Region, District, Municipality	Text	
00016	Coverage area	Geographical area where the sanitation system or service provides coverage or is legally obligated to serve	Geometry	Type: Sanitation system, Facility
00017	Sector	Name for a sub-region of the coverage area that is defined by geography or the physical layout of the collection network	Text	
00018	Route	Name or number of the service route, such as a scheduled emptying (desludging) route, which defines the assets assigned to a particular staff member or team	Text	Type: Sanitation point, Containment, Connection
00019	Address	Physical address or postal address where the asset is located; in the case of sewer or property connections, the address of the customer	Text	
00020	Urban-rural classification	Degree of urbanization, as defined by national standards or the Degree of Urbanisation method (OECD 2021)	Choice	Type: Sanitation system, Sanitation point
	<i>Urban</i>	Cities or densely populated areas		

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	<i>Peri urban</i>	Towns and semi-dense areas, suburbs, or peri urban areas surrounding cities		
	<i>Rural</i>	Regions with low population density		
00021	Community	Name of the community, town, or neighborhood where a sanitation system or facility is located	Text	Type: Sanitation system, Facility
00022	Accessible by road	Asset can be accessed by vehicles using a nearby road	Checkbox	
00023	Perimeter fence or wall installed	Fence or wall exists to provide physical security for the asset	Checkbox	
00025	Wheelchair accessible	Asset can be accessed and used by people using wheelchairs or with limited mobility	Checkbox	Type: Facility, Sanitation point, Structure
00026	Distance to nearest water source	Distance from the containment or dispersal structure to the nearest well, borehole, spring, or other groundwater source used for drinking water	Unit: Length	Type: Containment, Dispersal
00027	Land occupancy	Legal status of the land the asset occupies or crosses	Choice	All types except: Sanitation system, subsystem
	<i>Public or utility land</i>	Asset lies in public land, a road reserve, or land owned or controlled by the service provider		
	<i>Private land, with agreement</i>	Asset crosses land owned by others, with a documented easement, right-of-way, or written agreement		
	<i>Private land, without agreement</i>	Asset crosses land owned by others without a documented arrangement, such as informal or customary permission		
	<i>Unknown</i>	Land status has not been determined		
	Hierarchy			
00031	Level 1	Highest or most general level of categorization to which the asset belongs under the asset hierarchy	Text	
00032	Level 2	Second highest level of hierarchy	Text	
00033	Level 3	Third highest level of hierarchy	Text	
00034	Level 4	Fourth highest level of hierarchy	Text	
00035	Level 5	Fifth highest level of hierarchy	Text	
00036	Level 6	Sixth highest level of hierarchy	Text	
00037	Level 7	Lowest level of hierarchy	Text	

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Status and condition			
00041	Status	How the asset is currently being used or operated	Choice	Type is not Sanitation point
	<i>Active</i>	Asset is in use or ready to use		
	<i>Backup</i>	Asset is installed or available onsite to be used as a backup to other similar assets in case of maintenance or failure		
	<i>Not functional</i>	Asset is not currently operational due to a failure or maintenance problem		
	<i>Under construction</i>	Asset is not operational because it is under construction		
	<i>Decommissioned</i>	Asset is no longer in use because it is permanently out of service		
	<i>Disposed</i>	Asset has been sold, demolished, or relocated to a disposal site		
00042	Condition	The physical or structural integrity of the asset	Choice	
	<i>Excellent</i>	Designed to current standards, operable and well-maintained		
	<i>Good</i>	Showing some signs of minor wear and tear, with minimal impact on performance; or, operable and well-maintained but not designed to current standards		
	<i>Fair</i>	Structurally sound but some deterioration or damage that is beginning to impact performance or require more frequent maintenance		
	<i>Poor</i>	Functional but requires a high level of maintenance to continue operating or requires major rehabilitation or replacement		
	<i>Very poor</i>	Serious problems that impact performance or safety; major rehabilitation or replacement required in the short term		
00043	Functional status	Most recent observation of whether a sanitation system or sanitation point is working as designed	Choice	Type: Sanitation system, Sanitation point
	<i>Functional</i>	In good working condition and regularly providing service according to the specifications in the original design		
	<i>Partially functional</i>	Provides service on a regular basis (possibly in a reduced capacity) but repairs are needed due to some maintenance or structural issue		
	<i>Not functional</i>	No longer providing service on a regular basis; this could be due to maintenance issues, structural failure, or problems with physical access		

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	<i>Decommissioned</i>	Permanently taken out of service, abandoned, or could not be found		
	Financial			
00051	Current replacement cost (USD)	Cost (in US Dollars) that would be incurred to replace the asset at the current time; this value requires regular updating based on depreciation and current market conditions; also known as CRC	Number	
00052	Expected useful life	Number of years that the asset is estimated to function after installation if it is well maintained; also known as EUL	Number	
00053	Probability of failure	Measure of the likelihood of a failure happening at any point in time, based on an analysis of the potential failure modes of the asset	Choice	
	<i>Very low</i>			
	<i>Low</i>			
	<i>Moderate</i>			
	<i>High</i>			
	<i>Very high</i>			
00054	Consequence of failure	Measure of the direct and indirect costs of asset failure, which may include financial, environmental, and societal costs	Choice	
	<i>Very low</i>			
	<i>Low</i>			
	<i>Moderate</i>			
	<i>High</i>			
	<i>Very high</i>			
00055	Asset risk	Business or organizational risk of asset failure, usually calculated from the probability and consequence of failure	Choice	
	<i>Very low</i>			
	<i>Low</i>			
	<i>Moderate</i>			
	<i>High</i>			
	<i>Very high</i>			
00056	Original cost (USD)	Total price paid (in US Dollars) to initially acquire an asset	Number	

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
00057	Current rehabilitation cost (USD)	Cost (in US Dollars) that would be incurred to rehabilitate the asset at the current time	Number	
	Technical			
00061	Manufacturer	Name of the business that produced or sold the asset	Text	Type: Pump, Tank, Power, Meter, Electrical, Valve, Sensor, Analyzer, Vehicle, Connection, Datalogger, Treatment, Containment, Manhole
00062	Model	Manufacturer's designation for the particular model of the asset	Text	Type: Pump, Tank, Power, Meter, Electrical, Valve, Sensor, Analyzer, Vehicle, Connection, Datalogger, Treatment, Containment, Manhole
00063	Serial number	Unique identifying number or code assigned to a particular asset by the manufacturer or vendor, usually stamped or printed on the asset	Text	Type: Pump, Tank, Power, Meter, Electrical, Valve, Sensor, Analyzer, Vehicle, Connection, Datalogger, Treatment, Containment, Manhole
00064	Additional technical details	Any other technical information about the asset that is not captured by the attributes of this standard	Text	
00065	Installation date	Calendar date when the asset was installed or purchased, which begins the period of useful life	Date	
00066	Installed by	Name of the company or organization that originally installed or purchased the asset	Text	
00067	Funded by	Name of the funding authority, lender, or donor who funded the purchase of the system or facility	Text	Type: Sanitation system, Facility
00068	Rehabilitation date	Date of most recent work done to restore the asset to its original level of service and extend its life	Date	

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
00069	Last service date	Date of most recent service or maintenance performed; for containment assets, the date of most recent emptying	Date	
00075	Last calibration date	Calendar date when the device was last calibrated against a known reference or standard solution	Date	Type: Meter, Analyzer, Sensor, Datalogger
00070	Rehabilitated by	Name of the company or organization that performed the most recent rehabilitation of the asset	Text	
00071	Service interval	Recommended interval at which service or routine maintenance should be performed on the asset, such as the scheduled emptying frequency of a containment asset or the calibration of a measuring device	Unit: Time	All types except: Sanitation system, Subsystem, Facility, Connection
00072	Date of manufacture	Calendar date when the asset was manufactured	Date	Type: Pump, Tank, Power, Meter, Electrical, Valve, Sensor, Analyzer, Vehicle, Connection, Datalogger
00073	Decommissioning date	Calendar date when the asset was permanently taken out of service	Date	
00074	Owner	Name of the individual, organization, or authority that holds legal ownership of the asset, which may differ from the organization that operates or manages it	Text	
00076	Operator	Name of the individual, organization, or authority that operates or maintains the asset, if different from the owner; assets without an operator recorded are assumed to be operated by their parent system's operator	Text	Type: Sanitation system, Facility, Sanitation point, Vehicle
	Hydraulic			Type: Pump, Tank, Treatment, Meter, Valve, Flow control, Junction, Sampling point, Sanitation point, Containment, Manhole, Dispersal, Outfall, Pipe, Channel, Connection

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
00091	Upstream asset	Asset ID of the upstream asset from which this asset receives flow	Asset ID	
00097	Downstream asset	Asset ID of the downstream asset that receives flow from this asset	Asset ID	
00092	Nominal diameter	An approximate measurement of the internal diameter of a pipe, conduit, pump, or junction, which is used to classify according to standard sizes	Unit: Length	Type: Meter, Valve, Junction, Pipe, Connection
00093	Pipe length	Total length of the pipe itself, measured along its slope from the start to the end of the pipe and excluding manhole chambers, for use in hydraulic calculations and modeling	Unit: Length	Type: Pipe
00094	Material	Material of construction for pipes, junctions, chambers, and other conduits	Choice	Type: Meter, Valve, Junction, Pipe, Manhole, Containment, Channel
	<i>Cast iron</i>	Heavy brittle alloy of iron, carbon, and silicon that can be cast in specific shapes and has the ability to withstand shocks		
	<i>Ductile iron</i>	Similar material to cast iron, but with the primary graphite in a nodular or spheroidal form, giving it increased strength and flexibility		
	<i>Steel</i>	Iron-base alloy containing up to 2 percent carbon that is heat treated		
	<i>Stainless steel</i>	Chromium alloy steel that is resistant to rusting and corrosion		
	<i>Galvanized steel</i>	Steel that has a zinc coating to protect from corrosion and rust		
	<i>Vitrified clay</i>	Clay pipe fired at high temperature to form a chemically inert, corrosion-resistant material widely used in gravity sewers		
	<i>Concrete</i>	Mixture of cement and aggregate, usually reinforced with steel wire or rebar; includes reinforced concrete pipe and cast-in-place chambers		
	<i>Masonry</i>	Brick, stone, or concrete blocks that are bound together with mortar; common in legacy trunk sewers and chambers		
	<i>Asbestos cement</i>	Cement reinforced with asbestos fibers, found in legacy pipelines; subject to handling and disposal safety precautions		
	<i>Fiberglass (GRP)</i>	Reinforced plastic material composed of glass fibers embedded within a resin matrix; also known as glass-reinforced plastic		
	<i>PVC</i>	Polyvinyl chloride, a type of polymer frequently used in pipes and containers		
	<i>HDPE</i>	High-density polyethylene, a type of polymer used in gravity and pressure pipelines		

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	<i>Plastic</i>	Any high molecular weight polymer material that can be extruded, molded, or drawn into various shapes and sizes; use this category for any plastics that are not PVC or HDPE		
	<i>Other</i>	Any material not otherwise defined in this section		
00095	Minor loss coefficient	Dimensionless coefficient that can be used to account for the head loss associated with turbulence in bends, junctions, meters, and valves; usually applicable to pressurized (rising) mains	Number	Type: Junction, Valve, Meter
00096	Roughness coefficient	Dimensionless coefficient used to account for friction losses in flowing conduits, such as Manning's n for gravity sewers and open channels or Hazen-Williams C for pressurized mains	Number	Type: Pipe, Channel
00098	Conveyed content	Primary type of flow or material conveyed or handled by the asset	Choice	Type: Pump, Tank, Valve, Flow control, Meter, Junction, Sampling point, Manhole, Outfall, Pipe, Channel
	<i>Wastewater</i>	Untreated wastewater, including domestic sewage and blackwater		
	<i>Blackwater</i>	Wastewater from toilets containing excreta, with or without flushwater, conveyed separately from other streams		
	<i>Greywater</i>	Wastewater from washing, bathing, laundry, and food preparation that does not contain excreta		
	<i>Treated effluent</i>	Liquid stream that has undergone treatment prior to discharge or reuse		
	<i>Sludge or septage</i>	Sludge from treatment processes, or sludge and liquid mixtures emptied from on-site containment (faecal sludge)		
	<i>Combined sewage and stormwater</i>	Mixture of wastewater and stormwater conveyed in a combined sewer		
	<i>Stormwater</i>	Rainfall runoff only; assets dedicated solely to stormwater drainage are outside the scope of this standard		
	<i>Biogas</i>	Gas produced by anaerobic digestion, conveyed for recovery, use, or flaring		
	<i>Air</i>	Air conveyed for aeration, ventilation, or odor control		
	<i>Other</i>	Any other conveyed content not defined in this section		

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
00099	Invert level	Elevation of the lowest point of the internal surface of a channel or chamber, recorded against the same datum as Survey elevation	Unit: Length	Type: Manhole, Flow control, Outfall
	Points of contact			Type: Sanitation system, Subsystem, Facility, Sanitation point
00171	Primary contact name	Individual or office responsible for management or administration of the sanitation system or sanitation point	Text	
00172	Primary contact position	Title or role in the organization of the primary contact	Text	
00173	Primary contact phone number	Phone number of the primary contact	Text	
00174	Secondary contact name	Additional individual who may be contacted if the primary contact is not available	Text	
00175	Secondary contact position	Title or role of the secondary contact	Text	
00176	Secondary contact phone number	Phone number of the secondary contact	Text	
	Metadata			
00181	Date updated	Calendar date of the most recent update made to any attribute	Date	
00182	Date inspected	Calendar date of the most recent inspection or audit of the asset, as defined by organizational asset management policies	Date	
00183	Alternate ID	Optional additional identification code or number that is used by other data systems or organizations to track the asset	Text	
00184	Import code	Identification code for an asset that was assigned by an external database or asset management registry, used to maintain backwards traceability for assets that have been imported	Text	
00185	Permit or licence number	Identification number of the primary permit, licence, or registration issued for the asset by a regulator or government authority	Text	Type: Sanitation system, Facility, Sanitation point, Vehicle, Outfall

6.5 Type-specific attributes

The attributes defined in this section apply only to a specific asset type. Some asset types do not currently have type-specific attributes so they do not appear in this section.

6.5.1 System types

6.5.1.1 Sanitation system

Table 6. Sanitation system attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Sanitation system info			
10201	Sanitation system type	Categorization of the system by how excreta and wastewater are contained and conveyed: through a sewer network, in on-site containment, or both	Choice	
	<i>Sewered</i>	Piped sewer network that removes excreta and wastewater from the premises where they are produced and carries them to treatment or a discharge point		
	<i>Non-sewered (on-site)</i>	Excreta and wastewater are contained at or near the point of generation, with or without subsequent emptying, transport, and off-site treatment		
	<i>Mixed</i>	System serves users through both sewered and non-sewered services within its coverage area		
10202	Sewer network type	Categorization of the collecting network by what it is designed to carry: wastewater only in separate sewers, wastewater and stormwater together in combined sewers, or both	Choice	Sanitation system type: Sewered, Mixed
	<i>Separate</i>	Collecting network conveying primarily sanitary wastewater, with stormwater excluded or carried separately; also known as sanitary or foul sewers		
	<i>Combined</i>	Network that carries sanitary wastewater and also receives stormwater or snowmelt during wet weather		
	<i>Both separate and combined</i>	Collecting network contains both separate and combined sections		
10203	Management type	Type of organization that manages the operations, maintenance, and finances of the sanitation system	Choice	
	<i>Utility</i>	Organization created by government through legislative action or a private partnership or corporation with a legal mandate to provide sanitation services to a designated area		
	<i>Local government</i>	Local or municipal government office or staff directly manage the system		
	<i>Community-based organization</i>	Community-based organization, committee, or cooperative group of users that manages the system (also known as community-based management)		

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	<i>Sanitation authority</i>	Government agency or regulatory body established for the purpose of managing sanitation service provision		
	<i>Private</i>	Individual, partnership, or corporation that owns and operates a sanitation system but does not have a legal mandate to provide services to all users in an area		
	<i>Other</i>	Any other management entity not described by the other choices		
	<i>No management</i>	No individual or organization can be identified that manages the system		
10204	Number of households in service area	Total number of households within the area that a service provider is legally mandated to offer services to	Number	
10205	Number of households currently served	Number of households currently receiving service, whether through a functional sewer connection or through regular servicing of on-site containment (includes households with inactive, intermittent, or cut off service, if service could be re-established)	Number	
10206	System or network diagram	Photograph of a recent map or diagram showing the location of sanitation infrastructure, such as the collection network and treatment assets	Image	
10207	Total treatment capacity	Indicative amount of wastewater and faecal sludge that the system's treatment assets are able to receive and treat, within current resource and design constraints	Unit: Flow rate	
10208	Sewer technology	Predominant sewer technology used in the collecting network	Choice	Sanitation system type: Sewered, Mixed
	<i>Conventional gravity</i>	Gravity sewers laid at self-cleansing gradients and conventional depths, typically in public rights-of-way, with manhole access		
	<i>Simplified or condominial</i>	Smaller-diameter gravity sewers laid at shallower depths and reduced gradients, with simplified access chambers, often following shortened routes including through private property		
	<i>Settled (small-bore)</i>	Small-diameter sewers conveying settled, solids-free effluent from interceptor or septic tanks		
	<i>Pressure</i>	Sewers pressurized by pumps, such as grinder pump or septic tank effluent pump (STEP) systems		
	<i>Vacuum</i>	Network kept under negative pressure by a central vacuum station, with interface valves at connections		
	<i>Mixed</i>	Network combines more than one sewer technology		
10209	Design capacity (population equivalent)	Design capacity of the system expressed in population equivalent (p.e.), a standardized	Number	

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
		load unit where 1 p.e. is an organic biodegradable load with a five-day biochemical oxygen demand (BOD ₅) of 60 g of oxygen per day, independent of actual local per-capita loads (Directive (EU) 2024/3019, Art. 2(10))		

6.5.1.2 Subsystem

This section is reserved.

6.5.2 Facility types

6.5.2.1 Facility

Table 7. Facility attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Facility info			
20101	Facility type	Type of physical location where a specific function is performed in the collection, transport, treatment, or disposal/reuse of wastewater and faecal sludge	Choice	
	<i>Wastewater treatment plant</i>	Facility containing treatment works and process areas that treat sewered wastewater to an acceptable standard before discharge or reuse		
	<i>Faecal sludge treatment plant</i>	Facility designed to receive and treat faecal sludge delivered from on-site containment		
	<i>Pumping or lift station</i>	Structure containing pumps and related piping used to convey wastewater or faecal sludge		
	<i>Transfer station</i>	Facility where faecal sludge from small-scale collection is transferred to larger transport for onward conveyance		
	<i>Storage facility</i>	One or more tanks used to store wastewater, effluent, or faecal sludge		
	<i>Process area</i>	Area or building, often within a treatment plant, where a specific treatment process is carried out		
	<i>Retention or detention facility</i>	Basin or structure that temporarily holds flows, such as combined sewer overflow or equalization storage		
	<i>Other</i>	Any other type of facility not defined in this section		
20102	Process area type	Optional label used to categorize process areas within a facility	Text	Facility type: Process area
20103	Equipped to receive faecal sludge	The treatment facility is equipped to receive faecal sludge or septage delivered by	Checkbox	Facility type: Wastewater

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
		vehicle, in addition to sewer flow, such as through a discharge structure or receiving station		treatment plant

6.5.3 Vertical types

6.5.3.1 Sanitation point

Table 8. Sanitation point attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Sanitation point info			
31501	Sanitation point type	Type of technology used by the sanitation point to collect user excreta across sewer and non-sewered systems	Choice	
	<i>Flush/pour-flush toilet</i>	Toilet with a water seal, flushed by a cistern or by pouring water by hand, discharging to a sewer, septic tank, or pit		
	<i>Pit latrine with slab</i>	Dry toilet over a pit, covered by a durable, cleanable slab or platform with a drop hole or seat		
	<i>Pit latrine without slab or open pit</i>	Dry toilet over a pit lacking a durable slab, platform, or seat		
	<i>Ventilated improved pit (VIP) latrine</i>	Pit latrine fitted with a vent pipe and screen to reduce odors and insects		
	<i>Composting toilet</i>	Dry toilet in which carbon-rich material is added to excreta to produce compost, with or without urine separation		
	<i>Container-based toilet</i>	Toilet that collects excreta in a sealable, removable container regularly collected for treatment		
	<i>Bucket</i>	Bucket or other unsealed container used to collect excreta, periodically removed for disposal		
	<i>Hanging toilet or latrine</i>	Toilet built over a river, sea, or other water body into which excreta drops directly		
	<i>Urinal</i>	User interface that receives urine only, typically operated without a water seal		
	<i>Urine-diverting toilet</i>	Dry or flush toilet with a pan that separates urine from faeces at source for separate storage, treatment, or use		
	<i>Other</i>	Any other type of sanitation point not otherwise described in this section		
31505	Number of cubicles or stances	Total number of individual cubicles, stances, or seats available at the sanitation point	Number	
31507	Usage arrangement	Categorization of the sanitation point by who uses it and how access is arranged	Choice	

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	<i>Household (private)</i>	Used exclusively by members of a single household		
	<i>Shared (limited number of households)</i>	Used by a limited, known number of households, not open to the general public		
	<i>Communal or public</i>	Open to use by the general public or a community, not restricted to specific households		
	<i>Institutional</i>	Dedicated to a public-service institution such as a school, health facility, or market		
	<i>Other</i>	Any other usage arrangement not otherwise defined in this section		
31509	Separate facilities for women and girls	Sanitation point provides separate, private cubicles or blocks designated for use by women and girls	Checkbox	Usage arrangement: Shared (limited number of households), Communal or public, Institutional
31510	Pan style	Style of toilet pan or seat installed	Choice	
	<i>Squat</i>	Pan set into the floor, used in a squatting position		
	<i>Seated</i>	Pedestal or bench seat used in a sitting position		
	<i>Mixed</i>	Facility contains both squat and seated units		

6.5.3.2 Connection

Table 9. Connection attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Connection info			
32201	Connection type	Categorization of the connection based on the type of user or customer premises served	Choice	
	<i>Residential</i>	Connection serving one or more dwellings		
	<i>Commercial</i>	Connection serving a business or commercial premises		
	<i>Industrial</i>	Connection serving an industrial facility		
	<i>Institutional</i>	Connection serving a public-service institution such as a school, health facility, or government office		
	<i>Other</i>	Any other type of connection not described in this section		

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
32202	Customer ID	Identification code used by the billing or commercial system for the customer or property served by this connection	Text	
32203	Connection status	Current service status of the connection	Choice	
	<i>Active</i>	Connection is in service and discharging to the network		
	<i>Inactive</i>	Connection exists but is not currently in use, such as at a vacant property		
	<i>Disconnected</i>	Connection has been physically or administratively disconnected from the network		
	<i>Closed</i>	Customer account has been permanently closed		
32204	Interceptor tank installed	An interceptor tank, such as a septic tank, is installed upstream of this connection to retain settleable solids before discharge to the network (settled or solids-free sewerage)	Checkbox	
32205	Backflow prevention installed	A non-return valve or other backflow-prevention device is installed on the connection to prevent sewer surcharge from flowing back into the property	Checkbox	

6.5.3.3 Containment

Table 10. Containment attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Containment info			
31601	Containment type	Type of technology used to collect, store, and partially treat excreta and wastewater at or near the point of generation	Choice	
	<i>Septic tank</i>	Tank with two or more chambers separated by a baffle that settles and partially digests excreta before discharging effluent through an outlet		
	<i>Holding tank</i>	Fixed tank with no outlet, used to collect and store excreta or wastewater until emptied in place		
	<i>Pit</i>	Excavated chamber, lined or unlined, that collects and stores excreta without the settling-and-discharge design of a septic tank		
	<i>Twin pits or alternating pits</i>	Two pits used alternately so contents in the resting pit can decompose before emptying		
	<i>Composting chamber or dehydration vault</i>	Sealed above- or below-ground chamber that dries or composts excreta without water		
	<i>Container-based sanitation</i>	Sealable, removable containers regularly collected by a service provider and delivered		

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
		to treatment		
	<i>Biodigester</i>	Sealed anaerobic digestion tank that stabilizes excreta and produces biogas		
	<i>Other</i>	Any other type of containment not otherwise described in this section		
31602	Number of compartments	Number of separate chambers within the containment structure	Number	
31603	Storage volume	Effective storage or holding capacity of the containment structure	Unit: Volume	
31609	Containment depth	Vertical distance from the ground surface, slab, or access opening to the bottom of the containment structure; for pits, the excavated depth	Unit: Length	
31604	Lined or sealed	Degree to which the containment structure prevents infiltration of its contents into the surrounding soil	Choice	
	<i>Lined and sealed, watertight</i>	Structure is fully impermeable with no infiltration into surrounding soil		
	<i>Lined, impermeable walls, open bottom</i>	Walls are impermeable but the base allows liquid to infiltrate the soil		
	<i>Lined, semi-permeable walls, open bottom</i>	Walls allow some infiltration, such as honeycomb brick or open-jointed rings, and the base allows liquid to infiltrate the soil		
	<i>Unlined</i>	Walls and base are permeable, allowing liquid to infiltrate the surrounding soil		
	<i>Unknown</i>	Lining or sealing status has not been determined		
31605	Outlet arrangement	Categorization of where liquid effluent or overflow from the containment structure discharges	Choice	
	<i>Sealed, no outlet</i>	Fully enclosed structure with no effluent discharge; contents must be emptied		
	<i>To soak pit or leach field</i>	Effluent discharges to a soak pit or leach field for infiltration		
	<i>To sewer or open drain</i>	Effluent discharges to a sewer network or an open/storm drain		
	<i>To open ground or water body</i>	Effluent discharges uncontrolled to the ground surface or a water body		
	<i>Unknown</i>	Effluent destination has not been determined		
	<i>Other</i>	Any other outlet arrangement not otherwise defined in this section		
31606	Emptying access	Extent to which the containment structure can be physically accessed for emptying (desludging)	Choice	
	<i>Accessible by vacuum truck or mechanized equipment</i>	Site and compartment openings allow truck-mounted or motorized emptying equipment to reach and access the structure		

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	<i>Accessible by manual equipment only</i>	No vehicle access; contents can only be removed using handheld pumps or manual tools		
	<i>Not accessible for emptying</i>	No opening or access point exists to remove the structure's contents		
	<i>Unknown</i>	Emptying accessibility has not been assessed		
31607	Number of users sharing	Number of individual users that share use of this containment structure	Number	

6.5.3.4 Vehicle

Table 11. Vehicle attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Vehicle info			
31701	Vehicle type	Categorization of the mobile asset used to empty, transport, or transfer faecal sludge, septage, or wastewater, or to maintain the collection network	Choice	
	<i>Vacuum tanker</i>	Self-propelled truck fitted with an integrated vacuum pump and storage tank, used to suck and transport faecal sludge or septage		
	<i>Vacuum trailer (tractor-towed)</i>	Vacuum tank mounted on a towed trailer chassis and propelled by a separate tractor or vehicle rather than a self-contained truck		
	<i>Flatbed sludge truck</i>	Truck with an open or enclosed load bed used to transport drums, containers, or bins of faecal sludge without an integrated vacuum pump		
	<i>Motorized cart or small vehicle</i>	Small motorized unit, such as a tank mounted on a motorcycle, three-wheeler, or walking tractor		
	<i>Manual cart</i>	Human- or animal-powered cart used to transport containers or a tank of faecal sludge, sometimes fitted with a manual pump		
	<i>Sewer jetter (trailer or truck-mounted)</i>	High-pressure water jetting unit used to clear and clean sewer lines; trailer- or truck-mounted		
	<i>Other</i>	Any other vehicle type not described in this section		
31702	Registration number	Official license plate or registration number issued to the vehicle by a government authority	Text	
31703	Tank or load capacity	Maximum volume of faecal sludge, septage, or wastewater that the vehicle's tank or load bed can hold	Unit: Volume	

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
31704	Vacuum pump fitted	Whether the vehicle is equipped with a vacuum pump used to suction faecal sludge, septage, or wastewater from a containment system	Checkbox	

6.5.3.5 Manhole

Table 12. Manhole attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Manhole info			
31801	Manhole type	Categorization of the manhole by its access function	Choice	
	<i>Standard manhole</i>	Routine access chamber installed at changes in alignment, gradient, diameter, or at junctions		
	<i>Inspection chamber</i>	Small, shallow access chamber used in simplified or condominium sewerage in place of a full manhole		
	<i>Cleanout</i>	Pipe fitting extended to the surface allowing rodding or jetting access without a full chamber		
	<i>Drop manhole</i>	Manhole with an internal or external drop connection used where inflow and outflow inverts differ substantially		
	<i>Junction chamber</i>	Special-geometry chamber where multiple large sewers converge		
	<i>Catch basin or inlet (combined systems)</i>	Street-level inlet structure admitting stormwater to a combined sewer; dedicated stormwater drains are outside scope		
	<i>Other</i>	Any type of manhole not otherwise defined in this section		
31802	Chamber depth	Vertical distance from the manhole cover to the invert or floor of the chamber	Unit: Length	
31803	Cover shape	Shape of the manhole cover and frame	Choice	
	<i>Round</i>	Circular cover and frame		
	<i>Rectangular</i>	Rectangular or square cover and frame		
	<i>Other</i>	Any other cover shape not otherwise defined		
31804	Cover diameter	Nominal diameter of a round manhole cover	Unit: Length	Cover shape: Round
31807	Cover length	Length of a rectangular manhole cover, measured along its longer side	Unit: Length	Cover shape: Rectangular
31808	Cover width	Width of a rectangular manhole cover, measured along its shorter side	Unit: Length	Cover shape: Rectangular

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
31805	Cover load class	Load-bearing classification of the manhole cover per EN 124	Choice	
	<i>A15</i>	Areas usable only by pedestrians and cyclists, e.g. gardens		
	<i>B125</i>	Footways, pedestrian precincts and comparable areas		
	<i>C250</i>	Areas alongside kerbs, extending up to 0.5 m into the carriageway		
	<i>D400</i>	Carriageways of roads, including pedestrian streets and parking areas		
	<i>E600</i>	Areas subject to heavy wheel loads, e.g. ports and industrial pavements		
	<i>F900</i>	Areas subject to particularly heavy wheel loads, e.g. airport pavements		
	<i>Unknown</i>	Load class of the installed cover has not been determined		
31806	Access aid type	Type of fixed steps, ladder, or rungs installed inside the chamber for entry	Choice	
	<i>None</i>	No fixed steps, ladder, or rungs are installed		
	<i>Steps</i>	Individual fixed step irons or rungs installed in the chamber wall		
	<i>Ladder</i>	Continuous fixed ladder installed in the chamber		
	<i>Other</i>	Any other type of access aid not otherwise defined		

6.5.3.6 Junction

Table 13. Junction attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Junction info			
31001	Junction type	Type of fitting used where two or more pipes or conduits connect	Choice	
	<i>Wye</i>	Angled Y-shaped fitting joining a branch line to a main at a shallow angle to reduce solids deposition		
	<i>Tee</i>	Fitting in the shape of the letter T allowing a perpendicular three-way connection		
	<i>Bend</i>	Fitting that changes the direction of a pipe run, usually at a standard angle		
	<i>Reducer</i>	Fitting used to join two different sized pieces of pipe		
	<i>Saddle</i>	Fitting clamped or fused onto an existing main to create a branch connection		
	<i>Coupling or sleeve</i>	Fitting joining two pipe ends in line, including		

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
		repair couplings used in sewer rehabilitation		
	<i>Cap or plug</i>	Fitting closing a pipe end or stub, such as a capped connection awaiting future use		
	<i>Riser</i>	Vertical pipe section connecting a deep main upward toward a lateral or the surface		
	<i>Other</i>	Any type of junction not otherwise described in this section		

6.5.3.7 Valve

Table 14. Valve attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Valve info			
30801	Valve type	Purpose or design of a valve used to close off or regulate the flow of wastewater, sludge, or air in a pipe or closed conduit	Choice	
	<i>Gate</i>	Sliding gate that fully closes off or opens flow in a pipe		
	<i>Knife gate</i>	Thin-blade gate valve designed to cut through solids-laden or sludge flow without clogging		
	<i>Plug</i>	Rotating tapered or cylindrical plug used to shut off or throttle flow, tolerant of debris		
	<i>Check (non-return/flap)</i>	Allows flow in only one direction and closes automatically when flow tries to reverse		
	<i>Air release/vacuum</i>	Placed at a high point to release trapped air or admit air to prevent vacuum buildup in a pipe		
	<i>Backflow preventer</i>	Device that stops sewage from flowing back into a property or upstream section during surcharge		
	<i>Other</i>	Any other type of valve not otherwise described in this section		
30802	Date of last actuation	Most recent date that the valve was actuated to prevent degradation and verify functionality	Date	

6.5.3.8 Flow control

Table 15. Flow control attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Flow control info			
32101	Flow control element type	Physical form of the flow control element, independent of the function it serves	Choice	

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
		(32102)		
	<i>Side weir</i>	Weir set parallel to the direction of flow along the side of a channel or chamber, spilling excess flow over its crest once the water level rises above the sill		
	<i>Transverse weir</i>	Weir set across the direction of flow so that all flow above the sill level passes over the crest to the downstream or overflow side		
	<i>Leaping weir</i>	Opening or gap in the channel invert sized so that low dry-weather flow drops through to the intercepting sewer while higher storm flows leap across to the overflow side		
	<i>Movable weir</i>	Weir whose sill or crest can be raised, lowered, or tilted during operation to adjust the overflow level		
	<i>Gate or penstock</i>	Vertically sliding plate or sluice gate mounted at a channel, chamber, or tank opening, raised or lowered to close off or regulate free-surface flow		
	<i>Stop logs</i>	Removable beams or plates inserted into vertical guides to close off or throttle a channel or chamber opening; modern versions are aluminum stop plates		
	<i>Orifice or fixed throttle</i>	Fixed opening or plate, such as an orifice plate or throttle pipe, that limits the discharge passed downstream by restricting the flow area		
	<i>Vortex flow regulator</i>	Self-activating device with no moving parts in which rising head develops a swirling air core that sharply limits the discharge passed downstream		
	<i>Flow splitter</i>	Structure that divides incoming flow between two or more downstream lines or process units in set proportions, typically over fixed weirs sized in proportion to the receiving units' capacity		
	<i>Overflow pipe</i>	Relief pipe set at a fixed level in a tank, chamber, or storage volume that discharges excess flow once the water surface rises above its invert; a relief sewer conveying the overflow onward is registered as a Pipe asset		
	<i>Flume</i>	Open-channel constriction, such as a Parshall or Venturi flume, installed primarily to control velocity or level in a channel by forcing flow through critical depth; a flume whose primary purpose is flow measurement is registered as a Meter		
	<i>Other</i>	Any other flow control element not otherwise described in this section		
32102	Flow control function	Primary hydraulic function the flow control element serves in the system, independent of its physical form (32101)	Choice	

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	<i>Combined sewer overflow</i>	Diverts wet-weather flow exceeding the intercepting sewer's capacity from a combined sewer to a receiving water		
	<i>Storage or basin overflow</i>	Relieves a storage or equalization volume, such as a stormwater tank or retention basin, once its capacity is reached		
	<i>Process or clarifier overflow</i>	Controls the discharge of settled effluent over the outlet weirs of a clarifier or other treatment process unit		
	<i>Emergency overflow</i>	Provides a relief path that operates only under failure conditions, such as pump failure or blockage, to prevent uncontrolled flooding		
	<i>Flow splitting</i>	Divides flow in set proportions between parallel downstream units or lines		
	<i>Level control</i>	Maintains a set water level or velocity in an upstream channel, tank, or process unit		
	<i>Other</i>	Any other function not otherwise described in this section		

6.5.3.9 Structure

Table 16. Structure attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Structure info			
31401	Structure type	Type of ancillary structure present at a sanitation asset site	Choice	
	<i>Building</i>	Any enclosed structure designed to protect from the elements		
	<i>Perimeter</i>	Fence, wall, or other protective structure that encloses a land area		
	<i>Road</i>	Linear pathway that has been improved to allow for travel by vehicles and pedestrians		
	<i>Apron</i>	Smooth, impermeable surface constructed around a toilet, treatment unit, or vehicle discharge point to prevent spilled wastewater or sludge from pooling or infiltrating the ground		
	<i>Discharge structure</i>	Fixed structure, such as a receiving station, where vehicles discharge collected faecal sludge or septage into a treatment plant or transfer station		
	<i>Utility vault</i>	Underground room or vault providing access and protection to equipment		
	<i>Vehicle washing bay</i>	Hardstanding area with drainage where emptying and transport vehicles are washed down, typically at a faecal sludge receiving station or treatment plant		

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	<i>Flood protection bund</i>	Earthen or constructed barrier protecting a sanitation asset site against flooding		
	<i>Other</i>	Any type of structure not otherwise defined in this section		
	Perimeter data			
31411	Perimeter type	Type of perimeter design	Choice	Structure type: Perimeter
	<i>Fence</i>	Barrier constructed of vertical posts connected by horizontal sections of sturdy material		
	<i>Wall</i>	Permanent structure with a continuous surface		
	<i>Other</i>	Any type of perimeter structure not defined elsewhere in this section		
31412	Perimeter total length	Total length of the entire perimeter that is being defined as an asset	Unit: Length	Structure type: Perimeter
31413	Perimeter height	Typical or minimum height of the perimeter	Unit: Length	Structure type: Perimeter
31414	Perimeter setback distance	Typical or minimum distance from the perimeter from the structure or equipment it is designed to protect	Unit: Length	Structure type: Perimeter

6.5.3.10 Power

Table 17. Power attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Power info			
30401	Power type	Categorization of the power generating device based on design or principles of operation	Choice	
	<i>Grid</i>	Connection to an electrical transmission and distribution system		
	<i>Generator</i>	Diesel, gas, or petrol powered engine that converts rotary motion into electrical current		
	<i>Solar</i>	Solar or photovoltaic power system		
	<i>Wind</i>	Wind turbine or windmill equipped with an electrical generator		
	<i>Other</i>	Any other power type not defined in this section		
30402	Rated power output	Power output that can be sustained for long periods of time; for generators this is the continuous power rating	Unit: Power	
30403	Maximum power output	Power output that can be provided for short periods of time in response to surges in demand; for generators, sometimes referred to as the standby or intermittent power	Unit: Power	

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
		rating		
30404	Nominal voltage	Voltage assigned to a system or circuit of a given voltage class for the purpose of convenient designation; the actual operating voltage may vary above or below this value	Unit: Voltage	
30405	Rated voltage	Maximum voltage at which the power system can be operated within thermal and safety limits	Unit: Voltage	
30406	Maximum current	Maximum safe current, or starting current, that can be supplied for a short period of time	Unit: Current	
30407	Type of current	Mode of transmission for electrical current (AC/DC)	Choice	
	<i>Single phase AC</i>	Alternating current electrical system that uses two wires: one neutral wire and one phase wire in which the voltage rises to a peak in one direction and then reverses		
	<i>Three phase AC</i>	Alternating current electrical system that uses four wires: one neutral wire and three phase wires in which the voltage rises to a peak and reverses direction at different times		
	<i>DC</i>	Direct current system in which current always flows in one direction and the voltage does not change over time		
30408	Frequency of current	Frequency at which the voltage rises to a peak, reverses, and rises to a peak again	Choice	
	<i>60 Hz</i>	60 cycles per second		
	<i>50 Hz</i>	50 cycles per second		
	<i>Other</i>	Any non-standard current frequency		
	Generator data			
30421	Generator fuel	Type of fuel required by the generator	Choice	Power type: Generator
	<i>Gasoline or petrol</i>	Fuel made from crude oil and other petroleum liquids that is often used in internal combustion engines		
	<i>Diesel</i>	Distillate fuel oil that is used in compression ignition engines		
	<i>Other</i>	Any other fuel used in generators that is not described in this section		
30422	Generator apparent power	Product of the voltage and current provided in an AC power system, which does not account for the reactive power required by inductive loads in the circuit; sometimes referred to by its most common unit of measurement: kVA	Unit: Power	Power type: Generator
30423	Engine cooling system	Type of cooling system used by the generator engine	Choice	Power type: Generator

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	<i>Air</i>	Engine cooled by air circulation only		
	<i>Liquid</i>	Engine cooled by a circulating liquid, such as water or other coolant		
	Solar power data			
30424	PV panel type	Photovoltaic panel type	Choice	Power type: Solar
	<i>Polycrystalline</i>	Multicrystalline or poly panels, made of multiple pieces of silicon melted or joined together		
	<i>Monocrystalline</i>	Single crystalline or mono panel, made from a single silicon crystal		
	<i>Thin film</i>	Thin, flexible panel made from amorphous silicon or other materials		
	<i>Other</i>	Any other type of photovoltaic panel not described in this section		
30425	Total number of PV panels	Total number of photovoltaic panels installed in the power system	Number	Power type: Solar

6.5.3.11 Electrical

Table 18. Electrical attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Electrical info			
30701	Electrical type	Type of electrical asset	Choice	
	<i>Motor</i>	Electromechanical device for converting electrical energy into mechanical energy		
	<i>Actuator</i>	Device that uses electrical or pneumatic power to operate valves		
	<i>Power device</i>	Electrical or electronic device for managing the supply and distribution of electrical power		
	<i>Electronic</i>	Component or device that controls the flow of electrical currents for the purpose of information processing and system control		
	<i>Heater</i>	Device that converts electrical energy into heat		
	<i>Blower</i>	Device that uses electrical energy to move or pressurize air		
	<i>Other</i>	Any other type of electrical asset not defined in this section		
30702	Electrical power device type	Classification of electrical power device based on purpose or mode of operation	Choice	Electrical type: Power device
	<i>Control panel</i>	Controls electrical power distribution		

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	<i>Circuit breaker</i>	Automatic device for stopping the flow of current in a circuit in case of a fault		
	<i>Uninterruptible power supply (UPS)</i>	Energy storage device that ensures a constant supply of power, even during power outages		
	<i>Transformer</i>	Increases or decreases the voltage/current or vice versa		
	<i>Power conditioner</i>	Device that improves the quality of power delivered by smoothing out voltage spikes and reducing electrical noise		
	<i>Switch</i>	Device to disconnect or connect the conducting path in an electrical circuit		
	<i>Motor controller</i>	Device that controls the operation of an electric motor, such as a starter, soft starter, or variable frequency drive (VFD)		
	<i>Inverter</i>	Device for converting direct current into alternating current		

6.5.3.12 Meter

Table 19. Meter attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Meter info			
30601	Meter ID	Identification number or code used by the service provider to identify the meter	Text	
30602	Meter type	Location or function of the flow being measured within the wastewater, effluent, or sludge stream	Choice	
	<i>Influent</i>	Measures the flow of untreated wastewater or faecal sludge entering the sanitation system, a treatment plant, or a specific treatment unit		
	<i>Effluent</i>	Measures the discharge of treated effluent from a treatment plant or unit to disposal, typically for regulatory compliance		
	<i>Sludge or septage</i>	Measures flow of sludge, faecal sludge, or septage between treatment units or during transfer		
	<i>Reclaimed or reuse water</i>	Measures treated effluent supplied to a reuse customer or reclamation system, functioning like a customer water meter for billing and delivery		
	<i>Process or intermediate</i>	Measures flow between unit processes within a treatment train		
	<i>Other</i>	Any other metering function not otherwise described in this section		
30603	Meter design	Design or operating principle of the meter	Choice	

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	<i>Electromagnetic</i>	Measures the induced voltage generated by a conductive fluid, such as wastewater or sludge, flowing through a sensor		
	<i>Ultrasonic</i>	Measures flow velocity by transmitting and receiving sound waves through the flow path; tolerant of solids and grease		
	<i>Mechanical</i>	Turbine, displacement, or jet meter that measures flow using moving mechanical components; prone to fouling by solids		
	<i>Flume</i>	Shaped open channel, such as a Parshall flume, installed primarily to measure flow through a known relationship between depth and flow rate; a flume serving mainly control or relief functions is registered as a Flow control asset		
	<i>Weir with level sensor</i>	Weir paired with a level sensor that calculates flow rate from the measured head over the crest, installed primarily for measurement; a weir serving mainly overflow or control functions is registered as a Flow control asset		
	<i>Other</i>	Any other type of meter not otherwise described in this section		
30604	Minimum measurable flow rate	Lowest flow rate that can be measured by the meter	Unit: Flow rate	
30605	Meter measurement type	Type of measurement displayed by the meter (total volume, flow rate, or both)	Choice	
	<i>Total volume</i>	Totalizing meter, which displays only the total volume that has flowed through the meter		
	<i>Flow rate</i>	Instantaneous measurement of the current flow rate through the meter		
	<i>Combined</i>	Meter displays both the total volume and the instantaneous flow rate		

6.5.3.13 Sampling point

Table 20. Sampling point attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Sampling point info			
31101	Sampling point type	Location in the wastewater or faecal sludge stream where a sample is collected for water quality or process testing, including regulatory compliance points	Choice	
	<i>Influent</i>	Point where a sample of untreated wastewater or faecal sludge entering the system or a treatment plant is collected		
	<i>Effluent</i>	Point where a sample of treated effluent leaving a treatment plant or unit is collected,		

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
		often for regulatory compliance		
	<i>Process or intermediate</i>	Point between treatment units where a sample is collected to monitor process performance		
	<i>Sludge</i>	Point where a sample of sludge, faecal sludge, or septage is collected		
	<i>Receiving water</i>	Point in the water body receiving the discharge where a sample is collected to assess environmental impact		
	<i>Industrial discharge</i>	Point at an industrial user's connection where a sample is collected to verify pretreatment compliance		
	<i>Other</i>	Any type of sampling point not otherwise defined in this section		
31102	Regulatory compliance point	Sampling point designated for regulatory compliance monitoring under a discharge permit or national monitoring program	Checkbox	
31103	Autosampler installed	A fixed automatic sampler is installed at this point to collect composite or sequential samples	Checkbox	

6.5.3.14 Sensor

Table 21. Sensor attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Sensor info			
31201	Sensor type	Physical or field parameter measured by the sensor	Choice	
	<i>Level</i>	Height or depth of wastewater or sludge, measured from a known reference point		
	<i>Pressure</i>	Force pushing on a unit area		
	<i>Gas</i>	Presence or concentration of a hazardous or asphyxiant gas, such as hydrogen sulfide (H ₂ S), methane (CH ₄), or oxygen (O ₂), typically for confined-space monitoring in wet wells, manholes, or lift stations		
	<i>Rainfall</i>	Amount or intensity of precipitation, used to correlate infiltration/inflow or combined sewer overflow (CSO) events with rainfall		
	<i>Temperature</i>	Temperature of wastewater, sludge, or ambient air, used to monitor process conditions such as digestion		
	<i>Moisture</i>	Presence or absence of water in contact with the sensor		
	<i>Current</i>	Rate of flow of electric charge through a unit area		

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	<i>Voltage</i>	Electrical pressure available to cause a flow of current		
	<i>Other</i>	Any other parameter not otherwise defined in this section		
31202	Gas measured	Gas or gases that the sensor measures	Choice	Sensor type: Gas
	<i>Hydrogen sulfide (H2S)</i>	Toxic gas produced by anaerobic decomposition of sewage		
	<i>Methane (CH4)</i>	Flammable gas produced by anaerobic digestion of organic matter		
	<i>Oxygen (O2)</i>	Oxygen concentration, monitored for confined-space entry or aeration		
	<i>Ammonia (NH3)</i>	Toxic gas released from sewage and sludge handling		
	<i>Carbon monoxide (CO)</i>	Toxic combustion gas monitored in confined spaces		
	<i>Multi-gas</i>	Sensor measures several gases in one instrument		
	<i>Other</i>	Any other gas not otherwise described in this section		
31203	Gases measured	Gases that a multi-gas instrument measures, separated by commas — for example: hydrogen sulfide, methane, oxygen. Use the names given in Gas measured where they apply	Text	Gas measured: Multi-gas

6.5.3.15 Analyzer

Table 22. Analyzer attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Analyzer info			
31301	Analyzer type	Water quality or process parameter measured by the analyzer	Choice	
	<i>pH</i>	Measures the acidity or alkalinity of the stream		
	<i>Dissolved oxygen (DO)</i>	Measures dissolved oxygen concentration, such as in aeration basins		
	<i>Turbidity</i>	Measures suspended particulate cloudiness of the stream		
	<i>Ammonia</i>	Measures ammonia or ammonium concentration		
	<i>Nitrate</i>	Measures nitrate concentration, such as for denitrification control		
	<i>Total nitrogen or TKN</i>	Measures nitrogen in aggregate rather than as a single species: Kjeldahl nitrogen, which combines organic and ammonia nitrogen, or total nitrogen, which also includes nitrite and		

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
		nitrate; use Ammonia or Nitrate where a single species is monitored		
	<i>Phosphate or total phosphorus</i>	Measures phosphorus in the stream, either as reactive orthophosphate to control chemical or biological phosphorus removal, or as total phosphorus across all forms present for discharge compliance		
	<i>COD or BOD</i>	Measures chemical or biochemical oxygen demand of the stream		
	<i>Total organic carbon (TOC)</i>	Measures organically bound carbon in the stream, giving a same-shift indication of organic strength where the delay of a five-day BOD test is impractical		
	<i>Suspended solids</i>	Measures total suspended solids or mixed liquor solids concentration		
	<i>ORP</i>	Measures oxidation-reduction potential of the stream		
	<i>Chlorine</i>	Measures free or total chlorine residual, used to set the disinfection dose and to confirm that residual has been removed before effluent is discharged		
	<i>Conductivity</i>	Measures electrical conductivity of the stream		
	<i>Multiparameter</i>	Analyzer measures several parameters in one instrument		
	<i>Other</i>	Any other parameter not otherwise described in this section		
31302	Parameters measured	Parameters that a multiparameter analyzer measures, separated by commas	Text	Analyzer type: Multiparameter

6.5.3.16 Datalogger

Table 23. Datalogger attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Datalogger info			
32301	Transmission method	Method by which recorded data are transmitted from the datalogger to the systems or staff that use them	Choice	
	<i>None (manual download)</i>	Data are stored on the device and collected manually		
	<i>Cellular network</i>	Data are transmitted over a mobile network, such as GSM, GPRS, or LTE		
	<i>Radio</i>	Data are transmitted by a licensed or unlicensed radio link		
	<i>Satellite</i>	Data are transmitted via a satellite uplink		
	<i>Other</i>	Any other transmission method not described in this section		

6.5.3.17 Pump

Table 24. Pump attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Pump info			
30201	Pump type	Description of the pump mechanism or operating principle used to empty, convey, or transfer wastewater or faecal sludge	Choice	
	<i>Submersible non-clog/vortex</i>	Electrically powered pump with an open, recessed, or vortex impeller designed to pass solids without clogging; used in wet wells and lift stations		
	<i>Grinder</i>	Submersible pump fitted with a cutting or shredding mechanism that macerates solids before pumping; common in low-pressure sewer systems		
	<i>Screw (Archimedes)</i>	Inclined rotating screw that lifts wastewater or sludge within an open trough; tolerant of large solids and debris, common at headworks		
	<i>Progressive cavity (positive displacement)</i>	Rotor turning within a flexible stator that advances thick or viscous sludge at a constant flow rate regardless of pressure		
	<i>Diaphragm</i>	Motorized or manually operated pump that uses a flexible diaphragm to displace faecal sludge or wastewater; widely used for pit and septic tank emptying		
	<i>End-suction centrifugal</i>	Fluid enters along the pump axis, is accelerated by an impeller, and exits at a right angle to the shaft through a single suction port		
	<i>Axial flow</i>	Fluid is accelerated in the direction of the flow axis by an impeller running inside a casing; used for high-flow, low-head applications such as storm or combined sewer pump stations		
	<i>Manual pump</i>	Human-powered pump, such as the Gulper, MAPET, or manually operated diaphragm pump (MDHP), used to manually lift or convey faecal sludge		
	<i>Other</i>	Any other pump type not described in this section		
30202	Pump diameter	Maximum outer diameter of a submersible pump	Unit: Length	Pump type: Submersible non-clog/vortex
30203	Pump power rating	Power consumed by a pump operating under normal pressure and flow conditions, as defined by the manufacturer	Unit: Power	Pump type is not Manual pump
30204	Pump motor configuration	Whether the motor and pump should be tracked as a single asset or as separate assets; in cases where the motor is	Choice	Pump type is not Manual pump

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
		separate from the pump, this generally means that motors are interchangeable between pumps of a similar design		
	<i>Integrated with pump</i>	Motor and pump should be treated as the same asset		
	<i>Separate from pump</i>	Motor should be treated as a separate asset from the pump		
	Pump motor data			
30205	Pump motor voltage type	Type of electrical power required by the pump	Choice	Pump motor configuration: Integrated with pump
	<i>Single phase AC</i>	Alternating current electrical system that uses two wires: one neutral wire and one phase wire in which the voltage rises to a peak in one direction and then reverses		
	<i>Three phase AC</i>	Alternating current electrical system that uses four wires: one neutral wire and three phase wires in which the voltage rises to a peak and reverses direction at different times		
	<i>DC</i>	Direct current system in which current always flows in one direction and the voltage does not change over time		
30206	Pump motor voltage	Normal operating voltage of the pump, as recommended by the manufacturer	Unit: Voltage	Pump motor configuration: Integrated with pump
30207	Pump motor max current	Current drawn by the pump when operating at the maximum or full load power rating	Unit: Current	Pump motor configuration: Integrated with pump
30208	Pump motor max power	Maximum, or full load power rating of the pump, as recommended by the manufacturer	Unit: Power	Pump motor configuration: Integrated with pump
	Pump curve			
30221	Design flow	Flow rate of the pump required by the system design under normal operating conditions; should be within the acceptable flow range as defined by the pump curve	Unit: Flow rate	
30222	Design head	Head, or pressure, shown on the pump curve at the design flow rate	Unit: Pressure	
30223	Maximum flow (Q-max)	Maximum possible flow rate shown on the pump curve	Unit: Flow rate	
30224	Head at maximum flow (H-min)	Total head, or pressure, that corresponds to the maximum flow rate on the pump curve; also referred to as the minimum head	Unit: Pressure	
30225	Minimum flow (Q-min)	Minimum possible flow rate shown on the pump curve	Unit: Flow rate	

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
30226	Head at minimum flow (H-max)	Total head, or pressure, that corresponds to the minimum flow rate on the pump curve; also referred to as the maximum head	Unit: Pressure	
30227	NPSH-R	Net positive suction head required (or minimum) by the suction side of the pump in order to prevent cavitation and evaporation (flashing)	Unit: Pressure	

6.5.3.18 Tank

Table 25. Tank attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Tank info			
30301	Tank type	Function or purpose of the tank within the conveyance network or treatment facility	Choice	
	<i>Equalization or balancing tank</i>	Attenuates variation in flow or organic load to protect downstream treatment units		
	<i>Sludge or septage holding tank</i>	Temporarily stores sludge, septage, or faecal sludge before further treatment, transport, or disposal, within a treatment facility, transfer station, or conveyance network; an on-site tank storing excreta at the point of generation is a Containment asset (Holding tank)		
	<i>Chemical storage tank</i>	Stores coagulants, polymers, chlorine, or other treatment chemicals prior to dosing		
	<i>Effluent storage tank</i>	Stores treated effluent prior to discharge or reuse, such as for irrigation		
	<i>Wet well</i>	Collects wastewater or sludge ahead of a pump to provide pump suction volume and control cycling		
	<i>Portable tank</i>	Tank which can be moved from place to place; could be installed above or below ground		
	<i>Combined sewer overflow storage (tank or tunnel)</i>	Tank or deep tunnel that stores combined sewage during wet weather for later return to treatment		
	<i>Other</i>	Any other tank function not otherwise described in this section		
30302	Tank material	Main material of construction used in the tank	Choice	
	<i>PVC</i>	Polyvinyl chloride, a type of polymer frequently used in pipes and containers		
	<i>Plastic</i>	Any high molecular weight polymer material that can be extruded, molded, or drawn into various shapes and sizes; use this category for any plastics that are not PVC		

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	<i>Concrete</i>	Mixture of cement and aggregate, usually reinforced with steel wire or rebar		
	<i>Aluminum</i>	Lightweight metal composed mainly of elemental aluminum that may be alloyed with other elements		
	<i>Steel</i>	Iron-base alloy containing up to 2 percent carbon that is heat treated		
	<i>Stainless steel</i>	Chromium alloy steel that is resistant to rusting and corrosion		
	<i>Composite</i>	Structure that combines concrete with a steel support structure		
	<i>Fiberglass</i>	Reinforced plastic material composed of glass fibers embedded within a resin matrix		
	<i>Masonry</i>	Brick, stone, or concrete blocks that are bound together with mortar		
	<i>Other</i>	Any other type of tank construction not described in this section		
30303	Tank volume	Useful volume of the tank which can be supplied between the minimum and maximum operating levels	Unit: Volume	
30304	Tank shape	Basic shape or form factor of the tank	Choice	
	<i>Cylindrical</i>	Tank with a circular cross section		
	<i>Rectangular</i>	Tank with a rectangular cross section, including square		
	<i>Other</i>	Any other tank shape not described in this section		
	Tank dimensions			
30311	Tank diameter	Diameter of the base of a cylindrical tank	Unit: Length	Tank shape: Cylindrical
30312	Tank height	Distance from the bottom to the top of the tank, measured from outside	Unit: Length	
30313	Tank length	Length of the longest side of the base of a rectangular tank	Unit: Length	Tank shape: Rectangular
30314	Tank width	Length of the shortest side of the base of a rectangular tank	Unit: Length	Tank shape: Rectangular
30315	Height of tank bottom	Distance from the ground level to the bottom of the inside of the tank; negative values are used for depths below the ground	Unit: Length	
30316	Minimum operating level	Minimum fluid level allowed in the tank, measured from the tank bottom; this may be the height of the outlet or the minimum level allowed by a level control	Unit: Length	
30317	Maximum operating level	Maximum fluid level allowed in the tank, measured from the tank bottom; this may be the height of the overflow pipe or the maximum level allowed by a level control	Unit: Length	
	Tank details			

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
30323	Level indicator installed	Device is installed that provides a visual or electronic measurement of the tank fill level, used to trigger alarms or control pumps	Checkbox	

6.5.3.19 Treatment

Table 26. Treatment attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Treatment info			
30501	Treatment type	Categorization of treatment process by treatment method or objective, spanning liquid-stream (wastewater) and sludge/faecal sludge treatment technologies	Choice	
	<i>Screen</i>	Bar rack, mesh, or fine screen at the headworks that intercepts coarse solids, rags, and debris carried in the incoming flow		
	<i>Grit removal</i>	Channel, vortex unit, or aerated chamber at the headworks in which sand, gravel, and other dense inorganic particles settle out of the flow		
	<i>Grease or FOG removal</i>	Trap, skimming chamber, or flotation unit at the headworks that separates fats, oils, and grease from the flow		
	<i>Sedimentation or settler</i>	Tank or basin in which settleable solids separate from wastewater or faecal sludge under gravity, whether ahead of or following biological treatment		
	<i>Imhoff tank</i>	Two-story tank combining primary sedimentation with separate sludge digestion in a lower compartment		
	<i>Anaerobic baffled reactor (ABR)</i>	Series of baffled chambers that direct flow up and down through settled sludge for anaerobic treatment		
	<i>Anaerobic filter</i>	Fixed-bed reactor in which wastewater flows through a submerged filter medium supporting anaerobic biofilm		
	<i>Upflow anaerobic sludge blanket (UASB)</i>	Reactor in which wastewater flows upward through a blanket of anaerobic sludge granules for high-rate digestion		
	<i>Waste stabilization pond</i>	Shallow earthen pond that treats wastewater by sedimentation, digestion, and algal-bacterial processes; typically constructed in series as anaerobic, facultative, and maturation stages. Each pond in a series is registered as a separate Treatment asset, linked using upstream/downstream relationships.		
	<i>Aerated lagoon or pond</i>	Earthen basin in which mechanical or diffused aeration supplements natural		

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
		oxygenation to treat wastewater		
	<i>Free-water surface constructed wetland</i>	Engineered wetland in which wastewater flows as a shallow open layer above the soil, through the stems of emergent plants, resembling a natural marsh		
	<i>Horizontal subsurface flow constructed wetland</i>	Engineered wetland in which wastewater moves sideways through a planted bed of gravel or sand, staying below the surface of the media		
	<i>Vertical flow constructed wetland</i>	Engineered wetland in which wastewater is applied to the surface in intermittent doses and drains downward through a planted media bed to collection pipes at the base		
	<i>Trickling filter</i>	Fixed-film reactor in which wastewater trickles over a bed of rock or plastic media colonized by microorganisms		
	<i>Rotating biological contactor (RBC)</i>	Series of rotating discs partially submerged in wastewater that support a biofilm for aerobic treatment		
	<i>Aeration basin</i>	Basin in which wastewater is mixed with a suspended microbial culture and aerated to degrade organic matter		
	<i>Sequencing batch reactor (SBR)</i>	Single basin in which aeration, settling, and decanting of a suspended microbial culture occur in timed sequence rather than in separate units		
	<i>Membrane bioreactor (MBR)</i>	Basin holding a suspended microbial culture combined with membrane filtration in place of a separate settling stage		
	<i>Moving bed biofilm reactor (MBBR)</i>	Aerobic or anaerobic basin using free-floating plastic carriers that hold a biofilm in suspension		
	<i>Coagulation or flocculation</i>	Addition of chemicals with gentle mixing to destabilize and aggregate suspended particles for improved settling		
	<i>Filtration</i>	Granular, cloth, or membrane filter that removes residual particulates from treated effluent		
	<i>Chlorine contact tank</i>	Basin or vessel in which chlorine, in liquid, solid, or gaseous form, is contacted with effluent for disinfection		
	<i>UV or ozone disinfection unit</i>	Reactor or chamber that inactivates pathogens in effluent by non-chlorine means, such as ultraviolet light or ozone		
	<i>Chemical dosing system</i>	Storage, pump, and delivery equipment that doses chemicals for a treatment function such as pH control, odor control, or corrosion prevention		
	<i>Thickener</i>	Gravity or mechanical unit that concentrates sludge or faecal sludge solids before further treatment		

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	<i>Anaerobic digester</i>	Enclosed reactor in which microorganisms break down sludge or faecal sludge without oxygen, often producing biogas		
	<i>Unplanted drying bed</i>	Shallow bed with a filtration medium that dewateres sludge or faecal sludge by drainage and evaporation		
	<i>Planted drying bed</i>	Drying bed planted with wetland vegetation that enhances dewatering and stabilization of sludge or faecal sludge		
	<i>Dewatering unit</i>	Centrifuge, belt filter press, or screw press that mechanically separates liquid from sludge or faecal sludge		
	<i>Composting pad or windrow</i>	Pad, windrow, or vessel in which dewatered sludge or faecal sludge is mixed with organic bulking material and composted aerobically		
	<i>Lime stabilization</i>	Addition of lime to raise pH and reduce pathogens in sludge or faecal sludge		
	<i>Incineration or thermal treatment</i>	Combustion or other thermal process that reduces sludge or faecal sludge volume and destroys pathogens		
	<i>Aerobic digester</i>	Open or closed tank in which sludge is aerated to stabilize solids without biogas production; common in small and package plants		
	<i>Odor control (gas treatment)</i>	Equipment that contains, ventilates and treats odorous gases from collection or treatment assets		
	<i>Other</i>	Any other type of treatment process not already described in this section		
30502	Treatment level	Stage of treatment achieved by this unit, based on the pollutant-removal mechanism used; choices follow the legal definitions in Directive (EU) 2024/3019, Art. 2(11)-(14)	Choice	
	<i>Preliminary</i>	Removal of large solids, grit, or debris ahead of primary treatment; standard engineering practice		
	<i>Primary</i>	Physical or chemical process, such as settlement, removing a substantial portion of settleable solids and some organic load		
	<i>Secondary</i>	Process generally involving biological treatment that reduces biodegradable organic matter		
	<i>Tertiary</i>	Process that reduces nitrogen or phosphorus, or both		
	<i>Quaternary</i>	Process that reduces a broad spectrum of micropollutants		
	<i>Other</i>	Any treatment stage not otherwise described in this section		
30503	Design capacity of the unit	Design flow capacity of this specific treatment unit or process	Unit: Flow rate	

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
30504	Design organic load capacity of the unit	Design organic load capacity of this treatment unit, measured as the mass of five-day biochemical oxygen demand (BOD ₅) treated per day	Unit: Mass flow	
30505	Surface area	Plan surface area of the treatment unit at the water or bed surface, such as a pond, wetland, drying bed, or basin	Unit: Area	Treatment type: Waste stabilization pond, Aerated lagoon or pond, Free-water surface constructed wetland, Horizontal subsurface flow constructed wetland, Vertical flow constructed wetland, Unplanted drying bed, Planted drying bed
30506	Depth	Vertical distance from the water or bed surface to the bottom of the treatment unit; for ponds, the design liquid depth excluding freeboard	Unit: Length	Treatment type: Waste stabilization pond, Aerated lagoon or pond, Free-water surface constructed wetland, Horizontal subsurface flow constructed wetland, Vertical flow constructed wetland, Unplanted drying bed, Planted drying bed
30507	Liner	Type of liner installed to control seepage from the treatment unit into the surrounding soil	Choice	Treatment type: Waste stabilization pond, Aerated lagoon or pond, Free-water surface constructed wetland, Horizontal subsurface flow

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
				constructed wetland, Vertical flow constructed wetland, Unplanted drying bed, Planted drying bed
	<i>Geomembrane</i>	HDPE or other synthetic membrane liner		
	<i>Compacted clay</i>	Compacted clay or bentonite liner		
	<i>Concrete or masonry</i>	Concrete, masonry, or mortar lining		
	<i>Unlined</i>	Earthen, with no constructed liner		
	<i>Unknown</i>	Liner status has not been determined		
	<i>Other</i>	Any other liner type not otherwise defined in this section		
	Chemical data			
30521	Primary chemical used	Primary chemical stored and used by this treatment unit or process	Choice	Treatment type: Chlorination, Disinfection, Chemical feed, Coagulation or flocculation, Nutrient removal (nitrogen or phosphorus), Lime stabilization
	<i>Chlorine gas</i>	Gaseous molecular chlorine (Cl ₂) that forms hypochlorous acid and hypochlorite ion when dissolved in water		
	<i>Sodium hypochlorite (bleach)</i>	Solution in which sodium hypochlorite (NaOCl) is hydrolyzed in water to form hypochlorous acid (HOCl)		
	<i>Calcium hypochlorite</i>	Dry calcium hypochlorite (Ca(OCl) ₂) in powder or pellets dissolved in water to form hypochlorous acid		
	<i>Chloramine</i>	Disinfectant produced by mixing chlorine and ammonia, forming mainly monochloramine and dichloramine		
	<i>Chlorine dioxide</i>	Red-yellow gas that is very reactive and unstable, decomposing in water to form chlorite ion		
	<i>Sulfur dioxide</i>	Gaseous sulfur dioxide (SO ₂) used to dechlorinate effluent by neutralizing residual chlorine before discharge		

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	<i>Ozone (on-site generated)</i>	Ozone (O ₃) generated on site from air or oxygen for disinfection or oxidation; cannot be stored or transported in bulk		
	<i>Methanol</i>	Methanol (CH ₃ OH) dosed as a supplemental carbon source for biological denitrification; flammable liquid		
	<i>Ferric chloride</i>	Ferric chloride (FeCl ₃) solution used as a coagulant for solids and phosphorus removal; corrosive		
	<i>Alum</i>	Aluminum sulfate (Al ₂ (SO ₄) ₃) used as a coagulant for solids and phosphorus removal		
	<i>Lime</i>	Quicklime (CaO) or hydrated lime (Ca(OH) ₂) used for pH adjustment, chemical treatment, or sludge stabilization		
	<i>Caustic soda</i>	Sodium hydroxide (NaOH) solution used for pH adjustment or gas scrubbing; corrosive		
	<i>Acid</i>	Mineral acid, such as sulfuric or hydrochloric acid, used for pH adjustment; corrosive		
	<i>Polymer</i>	Organic polymer (polyelectrolyte) used as a flocculant or as a sludge conditioning and dewatering aid		
	<i>Other</i>	Any other primary chemical not described in this section		

6.5.3.20 Dispersal

Table 27. Dispersal attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Dispersal info			
31901	Dispersal type	Type of technology used to discharge treated or partially treated effluent to the soil for infiltration or evapotranspiration	Choice	
	<i>Soak pit</i>	Buried chamber with permeable walls through which pre-settled effluent seeps gradually into the ground		
	<i>Leach field or drainage field</i>	Buried perforated distribution pipes, bedded in gravel trenches, that spread effluent across an area of soil for infiltration		
	<i>Infiltration trench</i>	Narrow gravel- or sand-filled trench that receives effluent for subsurface infiltration		
	<i>Evapotranspiration bed</i>	Lined bed that disposes of effluent through evaporation and plant uptake rather than soil infiltration		
	<i>Mound system</i>	Raised soil mound used to disperse effluent where shallow soil or a high water table precludes a standard leach field		

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	<i>Other</i>	Any other type of dispersal structure not otherwise described in this section		

6.5.3.21 Outfall

Table 28. Outfall attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Outfall info			
32001	Outfall type	Categorization of the outfall by the type of discharge it conveys to the receiving environment	Choice	
	<i>Treatment plant outfall</i>	Point where treated effluent leaves a treatment facility to a receiving water body		
	<i>Combined sewer overflow (CSO)</i>	Engineered relief point on a combined network that diverts wet-weather flow exceeding downstream capacity to a receiving water; the weir, gate, or other element performing the diversion inside the chamber is registered separately as a Flow control asset		
	<i>Emergency overflow</i>	Point designed to relieve the system only under abnormal or emergency conditions; the element providing the relief path is registered separately as a Flow control asset		
	<i>Other</i>	Any type of outfall not otherwise defined in this section		
32002	Receiving water name	Name of the water body receiving the effluent or overflow, if one exists	Text	
32003	Screen or control installed	A physical screen, grate, tide gate, or flow-control device is installed at the outfall to remove solids or regulate discharge	Checkbox	
32004	Receiving environment type	Type of water body or environment receiving the discharge	Choice	
	<i>River or stream</i>	Flowing inland surface water		
	<i>Lake or reservoir</i>	Standing inland surface water body		
	<i>Estuary</i>	Transitional water where a river meets the sea		
	<i>Coastal water</i>	Marine water along the coast		
	<i>Agricultural land or reuse system</i>	Effluent is delivered to agricultural land, an irrigation network, or another planned reuse system rather than discharged to a water body		
	<i>Other</i>	Any other receiving environment not defined in this section		

6.5.3.22 Other vertical

Table 29. Other vertical attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
39901	Other vertical asset type	Description or categorization of the other type of vertical asset	Text	

6.5.4 Horizontal types

6.5.4.1 Pipe

Table 30. Pipe attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Pipe info			
40101	Pipe type	Category of sewer pipe by its position and function in the collection network	Choice	
	<i>Lateral (property connection)</i>	Small-diameter sewer connecting a single building or premises to the public network; the point where wastewater enters the collection system		
	<i>Condominial branch</i>	Shared small-diameter sewer that collects wastewater from a group of properties, often routed through private lots, sidewalks, or backyards, and conveys it to a main; the defining element of simplified or condominium sewer systems		
	<i>Main</i>	Sewer that aggregates flow from laterals and carries it toward a trunk sewer or interceptor		
	<i>Trunk</i>	High-capacity sewer receiving flow from mains and carrying it toward an interceptor or a treatment facility		
	<i>Interceptor</i>	Highest-order sewer in a network; collects flow from trunk and main sewers and delivers it to a treatment facility		
	<i>Rising main (force main)</i>	Pressurized pipe used to convey wastewater, typically downstream of a pump station		
	<i>Inverted siphon</i>	Depressed pipe or set of parallel pipes passing under an obstacle while maintaining self-cleansing velocity		
	<i>Other</i>	Any other type of sewer pipe not defined in this section		
40103	Nominal pressure	Numerical designation of design pressure that is a convenient round number for reference purposes	Unit: Pressure	Pipe type: Rising main (force main), Inverted siphon

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
40104	Pipe depth below ground	Depth of the pipe below the local ground surface	Unit: Length	
40105	Upstream invert level	Elevation of the lowest point of the internal pipe surface at the upstream end, recorded against the same datum as Survey elevation	Unit: Length	
40106	Downstream invert level	Elevation of the lowest point of the internal pipe surface at the downstream end, recorded against the same datum as Survey elevation	Unit: Length	
40107	Cross-section shape	Shape of the pipe's internal cross-section	Choice	
	<i>Circular</i>	Round pipe cross-section		
	<i>Egg-shaped</i>	Ovoid profile, narrower at the invert, used to maintain velocity at low flows		
	<i>Rectangular</i>	Rectangular closed profile, typically for large box sewers		
	<i>Trapezoidal</i>	Trapezoidal profile, typically for large culverts or box sewers		
	<i>U-shaped</i>	Profile with a rounded (semi-circular) invert and straight vertical side walls		
	<i>Other</i>	Any other cross-section shape not otherwise defined		
40108	Gradient	Slope of the pipe between upstream and downstream inverts, expressed as a percentage	Number	

6.5.4.2 Channel

Table 31. Channel attributes

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
	Channel info			
40201	Channel cross-section shape	Shape of the open channel's cross-section	Choice	
	<i>Rectangular</i>	Rectangular open channel profile		
	<i>Trapezoidal</i>	Trapezoidal open channel profile, common for larger constructed drains		
	<i>Circular</i>	Circular open channel profile		
	<i>U-shaped</i>	Profile with a rounded (semi-circular) invert and straight vertical side walls		
	<i>Irregular or natural</i>	Unformed earthen or naturally shaped channel cross-section		
	<i>Other</i>	Any other channel cross-section shape not otherwise defined		
40202	Lined	The channel bed and sides have a constructed lining, such as concrete,	Checkbox	

Code	Group, Attribute, or Choice Name	Description	Data type	Conditions
		masonry, or a synthetic membrane, to control seepage and erosion, as opposed to an unlined earthen channel		
40203	Covered or open	Whether the channel is open to the atmosphere or has a fixed cover or slab over it	Choice	
	<i>Open</i>	Channel is open to the atmosphere along its length		
	<i>Covered</i>	Channel has a fixed cover or slab over it		
40204	Channel depth	Vertical distance from the top of the channel side or lining to the invert (lowest point) of the channel, measured internally	Unit: Length	
40205	Channel top width	Internal width of the channel measured at the top, between the inner faces of the channel sides	Unit: Length	
40206	Gradient	Slope of the channel between its upstream and downstream inverts, expressed as a percentage	Number	

7 Bibliography

- African Ministers' Council on Water (AMCOW). (2021). *African sanitation policy guidelines*. AMCOW. ISBN 978-978-990-423-5. <https://amcow-online.org/african-sanitation-policy-guidelines-aspg/>
- American Water Works Association Asset Management Committee. (2018). *AWWA asset management definitions guidebook (Version 1.0)*. American Water Works Association.
- Athena Infonomics, World Bank, & Bill & Melinda Gates Foundation. (2021). *Citywide Inclusive Sanitation — Monitoring, Learning & Evaluation (CWIS-MLE): CWIS framework, summary indicators and links*. <https://www.cwiscities.com>
- BIPM. (2019). *Le Système international d'unités / The International System of Units ('The SI Brochure') (9th ed.)*. Bureau international des poids et mesures. <https://www.bipm.org/en/publications/si-brochure>
- Bremen Overseas Research and Development Association (BORDA). (2017). *DEWATS implementation by BORDA: Statistics on DEWATS implementation 1989–2017* [Brochure].
- Bundesministerium für Wohnen, Stadtentwicklung und Bauwesen (BMWSB) & Bundesministerium der Verteidigung (BMVg). (2025). *Baufachliche Richtlinien Abwasser (BFR Abwasser), Stand Januar 2025* [incl. ISYBAU-Austauschformate Abwasser (XML), version 2024; managed by Niedersächsisches Landesamt für Bau und Liegenschaften]. <https://www.bfr-abwasser.de>
- Butcher, K. S., Crown, L. D., & Gentry, E. J. (2006). *The international system of units (SI) — Conversion factors for general use (NIST SP 1038)*. National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.SP.1038>
- Davis, M. L. (2010). *Water and wastewater engineering: Design principles and practice*. McGraw-Hill.
- Directive (EU) 2024/3019 of the European Parliament and of the Council of 27 November 2024 concerning urban wastewater treatment (recast). (2024). *Official Journal of the European Union*, L series, 2024/3019, 12 December 2024.
- Direktorat Pengembangan Kawasan Permukiman, Direktorat Jenderal Cipta Karya, Kementerian Pekerjaan Umum dan Perumahan Rakyat. (2022). *Buku saku sanitasi* [Sanitation pocket book]. Government of Indonesia.
- Esri. (n.d.). *Combined Sewer Utility Network Essentials data dictionary* (ArcGIS Solutions 12.1 reference). https://doc.arcgis.com/en/arcgis-solutions/12.1/reference/other/CombinedSewerEssentials_UtilityNetwork.html
- Esri. (n.d.). *Sewer Utility Network Essentials data dictionary* (ArcGIS Solutions 12.1 reference). https://doc.arcgis.com/en/arcgis-solutions/12.1/reference/other/SewerEssentials_UtilityNetwork.html
- Esri. (n.d.). *Sewer Utility Network Expanded data dictionary* (ArcGIS Solutions 12.1 reference). https://doc.arcgis.com/en/arcgis-solutions/12.1/reference/other/SewerExpanded_UtilityNetwork.html

- European Commission, INSPIRE Temporary MIWP 2021–2024 sub-group 2.3.1. (2024). *D2.8.III.6 Data specification on utility and government services — Technical guidelines* (published 31 January 2024). INSPIRE — Infrastructure for Spatial Information in Europe.
- European Environment Agency. (2026). *UWWTD Article 15 reporting — Data dictionary* (Dataset 3569; released 17 April 2026; reference year 2024) [Eionet Data Dictionary, Reportnet]. <https://dd.eionet.europa.eu/datasets/latest/UWWTDArt15>
- European Committee for Standardization. (2015). *Gully tops and manhole tops for vehicular and pedestrian areas — Part 1: Definitions, classification, general principles of design, performance requirements and test methods (EN 124-1:2015)*.
- Government of Zimbabwe. (2013). *Sanitation focused participatory health and hygiene education: Training of trainers field guide* (revised ed.).
- Government of Zimbabwe. (2026). *The Zimbabwe national sanitation and hygiene strategy 2026–2030* (final, 11 March 2026).
- Gutterer, B., Sasse, L., Panzerbieter, T., & Reckerzügel, T. (2009). *Decentralised wastewater treatment systems (DEWATS) and sanitation in developing countries: A practical guide* (A. Ulrich, S. Reuter, & B. Gutterer, Eds.). BORDA; WEDC, Loughborough University.
- Inter-American Development Bank. (n.d.). *RedBasica* [Computer software — sewerage network design tool, EL-BID open-source programme]. GitHub. https://github.com/EL-BID/red_basica
- International Organization for Standardization. (2014). *Asset management — Management systems — Guidelines for the application of ISO 55001 (ISO 55002:2014)*.
- International Organization for Standardization. (2014). *Asset management — Management systems — Requirements (ISO 55001:2014)*.
- International Organization for Standardization. (2014). *Asset management — Overview, principles and terminology (ISO 55000:2014)*
- International Organization for Standardization. (2016). *Guidelines for the management of assets of water supply and wastewater systems — Part 1: Drinking water distribution networks (ISO 24516-1:2016)*.
- International Organization for Standardization. (2017). *Guidelines for the management of assets of water supply and wastewater systems — Part 3: Wastewater collection networks (ISO 24516-3:2017)*.
- International Organization for Standardization. (2019). *Asset management — Guidance on the alignment of financial and non-financial functions in asset management (ISO/TS 55010:2019)*.
- International Organization for Standardization. (2019). *Guidelines for the management of assets of water supply and wastewater systems — Part 4: Wastewater treatment plants, sludge treatment facilities, pumping stations, retention and detention facilities (ISO 24516-4:2019)*.
- Kementerian Pekerjaan Umum dan Perumahan Rakyat. (2017). Peraturan Menteri PUPR Nomor 04/PRT/M/2017 tentang penyelenggaraan sistem pengelolaan air limbah domestik [Ministerial Regulation No. 4/2017 on domestic wastewater management systems (SPALD)]. Government of Indonesia.
- Mara, D. (2004). *Domestic wastewater treatment in developing countries*. Earthscan.

- Metcalf & Eddy, Inc./AECOM. (2014). *Wastewater engineering: Treatment and resource recovery* (5th ed.; G. Tchobanoglous, H. D. Stensel, R. Tsuchihashi, & F. L. Burton, Eds.). McGraw-Hill Education.
- Ministry of Health and Child Care, Government of Zimbabwe. (2024). *The Zimbabwe national sanitation and hygiene policy*.
- mWater Foundation. (2022). The water system management standard: Part 1 — Asset management (Version 1.0, 7 April 2022). <https://www.mwater.co/blog/asset-systems>
- OECD, European Commission, Food and Agriculture Organization of the United Nations, United Nations Human Settlements Programme, International Labour Organization, & The World Bank. (2021). *Applying the Degree of Urbanisation: A methodological manual to define cities, towns and rural areas for international comparisons*. OECD Publishing. <https://doi.org/10.1787/4bc1c502-en>
- SFD Promotion Initiative. (2018). *SFD manual — Volume 1 and 2* (Version 2.0). Sustainable Sanitation Alliance. <https://sfd.susana.org>
- Stichting RIONED. (2024). *Gegevenswoordenboek Stedelijk Water (GWSW)* (Version 1.6.1, 1 August 2024) [OWL-2/RDF ontology and CSV downloads]. <https://data.gwswnl>
- Strande, L., Ronteltap, M., & Brdjanovic, D. (Eds.). (2014). *Faecal sludge management: Systems approach for implementation and operation*. IWA Publishing. <https://doi.org/10.2166/9781780404738>
- TEKSI Association. (2025). *TEKSI Wastewater: Open-source reference implementation of VSA-DSS 2020 and SIA 405* [PostgreSQL/PostGIS data model; successor to QGEP]. GitHub. <https://github.com/tekxi/wastewater>
- Tilley, E., Ulrich, L., Lüthi, C., Reymond, P., Schertenleib, R., & Zurbrugg, C. (2014). *Compendium of sanitation systems and technologies* (2nd rev. ed.). Eawag — Swiss Federal Institute of Aquatic Science and Technology.
- World Health Organization & United Nations Children's Fund. (2022). *Guidance for monitoring safely managed on-site sanitation (SMOSS): Draft prepared for Phase 2 pilots* (August 2022). WHO/UNICEF Joint Monitoring Programme for Water Supply, Sanitation and Hygiene. <https://washdata.org/monitoring/sanitation/safely-managed-on-site-sanitation>
- World Health Organization. (2018). *Guidelines on sanitation and health*. WHO.